

GCSS-MC
POST DEPLOYMENT SYSTEM SUPPORT (PDSS)
PERFORMANCE WORK STATEMENT (PWS)

15 August 2024

Section C - Descriptions and Specifications

PERFORMANCE WORK STATEMENT

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1 OVERVIEW

1.1 SCOPE

This Performance Work Statement (PWS) sets forth the requirements for the Offeror to provide Post Deployment System Support (PDSS) for the Global Combat Support System – Marine Corps/Logistics Chain Management (GCSS-MC/LCM) Increment 1 system, migration of Production and Production Support System (PSS) environments to Cloud One Infrastructure as a Service (IaaS), sustainment of GCSS-MC in IaaS, and decommissioning of GCSS-MC. The scope of support includes: Service Management, Service Operations (Enterprise Service Desk and GCSS-MC Operations Center (GOC) support for the GCSS-MC/LCM Increment 1 production system); Production and Pre-Production system sustainment; Solution Development Environment; Enterprise Training & Training Devices; Product lifecycle support; Service Transition for Change Requests (CRs) and potential tasks to support GCSS-MC future initiatives. The Offeror shall follow established DoD, systems engineering, software engineering, and program processes, policies and regulations in executing defined tasks. Not all tasks found in the base contract's PWS will necessarily be required or executed during a single Task Order. Successful Performance of this PWS and subsequent Task Orders will require interfacing with a multitude of personnel from multiple Government and contractor agencies, such as, but not limited to the GCSS-MC Program Management Office (PMO); Marine Corps Cyber Operations Group (MCCOG); and other vendors and Contractors as identified by the Government.

1.2 PROGRAM BACKGROUND

GCSS-MC/LCM Family of Systems (FoS) is the technology centerpiece for logistics modernization in the Marine Corps. It provides accurate, near real time integrated logistics information and enterprise-wide visibility of logistics data, enabling Marines to make informed decisions about the logistics chain commensurate with the operational tempo. GCSS-MC/LCM Increment 1 began full sustainment in December 2015.

The term "system" will be used interchangeably with GCSS-MC/LCM Increment 1 throughout the body of this document to refer to GCSS-MC/LCM Increment 1, inclusive of its software, hardware, firmware, and hosting environment. The term "application" refers to the Oracle Enterprise Business Suite (EBS) and other software components of the system. PDSS activities required to support the system are consistent with Defense Acquisition System Operations and Support activities, Milestone C Phase, and are listed in the following table:

System and Database Administration	System Integration
Interface & Data Management	Application Sustainment/Development
Information Assurance/Cybersecurity	Configuration/Change Management
Defect Management	Engineering Change Proposals
Asset Tracking	Training & Training Devices Management

Table 1: GCSS-MC/LCM Increment 1 PDSS Activities

GCSS-MC/LCM Increment 1 provides the Marine Air-Ground Task Force (MAGTF) and Supporting Establishment with an integrated, distributed logistics capability to plan, manage, execute, and monitor logistics business processes and sub-processes as depicted in Figure 1 below. Specific functions include asset visibility, supply, maintenance, readiness, task organization, request tracking, and financial information. It supports over 22,000 enterprise users from over 75 worldwide locations in both garrison and deployed environments. GCSS-MC/LCM Increment 1 is the first system in the portfolio and serves as the starting point for addressing several key shortfalls impacting the

logistics community, including: multiple entry points for submitting and managing requests, a lack of enterprise asset visibility, inefficient capacity management and enterprise-wide standard processes, and the inability to monitor asset availability and maintain situational awareness. Figure 1 depicts current logistics business processes and sub-processes supported by GCSS-MC/LCM Increment 1.

Business Process	Business Sub Process	Business Process	Business Sub Process
Request Management	Create or Update Service Request	Maintenance	Field Maintenance with Replacement Parts
	Request for Service		Create Preventative Maintenance Schedule
	Mass Transfer of Install Base Ownership		Readiness
	Retire Install Base Items		Repair with Tasks
	Maintenance-Related Returns		Create Warranty Contract
	Request for Supply		Discrete Manufacturing
	Request for Technical Assistance		OBIEE Reports
	Create and Maintain Accountable and Responsible Officers	Finance	Create Budget
	Flagging Install Base Items for Readiness		Depreciate Assets
	OBIEE Reports		Automated Processes
Supply	Order Management Execution		Budget Inquiry
	Secondary Repairable or T/E Items		Perform Period End Account Maintenance
	Advance Supply Chain Planning		Year End Close
	Miscellaneous Inventory Transactions		Funds Check Internal
	Classify/Declassify SECREP		ERS Invoices Created and Matched to Purchase Orders
	Requisition to Receipt		New Year Open
			General Ledger Journal and Account Inquiry
	Demand Planning		Period End Processing and Reporting
	Spares Management Minimum/Maximum Planning		Update Value of Fixed Assets
	Item Master and Bill of Materials (BOM)		OBIEE Reports
	Inquiries and Reports	System Administration	Task Organization
	Forecasting		Daily Business Intelligence
	Inventory Minimum/Maximum Planning		Reports
	Add New System Item		User Management
	Readiness		Tools
	Return of Spares		Login
	Supply Costing		Logout
	Quality		
	Warehouse Management		

Figure 1: GCSS-MC/LCM Increment 1 Logistics Business Processes & Sub-Processes

In 2015, GCSS-MC/LCM Increment 1 fielded Release 1.1.1, providing an interim deployable capability for GCSS-MC users either preparing to operate or operating in forward deployed environments. Release 1.1.1 includes an Enterprise Automated Task Organization (EATO) capability, Mobile Field Service (MFS) technology, and integration of a Tactical Wide Area Network (WAN) optimization solution. EATO provides Marines the ability to create task-organized units that can execute logistics chain management functions. MFS provides a GCSS-MC user the ability to perform limited supply and maintenance functions in the absence of a network connection. However, this functionality is not currently being utilized. The Tactical WAN optimization solution is comprised of a suite of Commercial Off-the-Shelf (COTS) Riverbed Steelhead Appliances and Steelhead Mobile Controller hardware that is integrated into the GCSS-MC/LCM system architecture to optimize deployed GCSS-MC user sessions and reduce latency caused by poor tactical network connections. In addition, GCSS-MC/LCM Increment 1 was configured to support advanced warehouse management functions. This capability will be fully fielded in 2QFY19 and the Offeror will be responsible for the sustainment after fielding.

1.3 SYSTEM DESCRIPTION

The GCSS-MC/LCM Increment 1 system is an integrated composite Enterprise Resource Planning (ERP) system with the Oracle EBS product at its core. Components of this integrated system include a customized Common Access Card (CAC) enabled authentication architecture with role based authorization, a reporting capability handled by the Oracle Business Intelligence Enterprise Edition (OBIEE); an Advanced Planning Suite (APS); a MFS application; and customized workflows in support of 20+ external partners via 40+ interfaces utilizing the Oracle Service Oriented Architecture (SOA) tools with Business to Business (B2B) functionality. GCSS-MC/LCM increment 1 is currently comprised of 3 “instances” that consist of six environments serving different purposes. Enterprise Production System (PROD), Production Support System (PSS), and Solution Development Environment (SDE) environments are located in Cloud One. The Solution Development Environment (SDE) instance contains the Development (DEV), Configuration Management (CM) and Training environments. The Mobile Training Suites (MTS) instance is housed within its own environment. The Training environment, although it can be hosted on shared hardware, is a separate environment that allows for the development, testing, and dissemination of training system updates and training material. The MTSs are utilized by the Formal Learning Centers (FLCs) and each of the major commands to conduct sustainment training of the GCSS-MC/LCM Increment 1 system. Each instance has its own environment of the GCSS-MC/LCM Increment 1 application to support its function. The following subparagraphs provide further description of the GCSS-MC environments.

Enterprise Production System

The Enterprise Production System, commonly referred to as PROD, is comprised of Oracle Engineered Hardware (OEHW) with the Exadata 6-2 as its foundation. The PROD is located in the Cloud One IaaS environment. The GCSS-MC/LCM Increment 1 Production system is web-accessible to over 25,000 worldwide users, serving between 5,000 and 6,600 distinct users per day, with a daily maximum concurrent usage of 1,200 to 1,650 users.

Production Support System

The PSS resides in the same environment as the PROD but is logically separated from the Production environment. The PSS is used to conduct Government Acceptance Testing (GAT) and the staging of changes before they are deployed to the Production system. The term “Production” or “Production System” used in this solicitation shall refer to the Enterprise Production System and the Production Support System unless otherwise stated.

Solution Development Environment

The GCSS-MC/LCM SDE resides in Cloud One IaaS environment. The SDE system mirrors the R19 Production environment. The SDE provides for the development, testing, troubleshooting and CM of the GCSS-MC R19 application. An SDE consists of the DEV and CM environments and has the ability to include additional concurrent environments. The DEV provides an environment for application development, troubleshooting, and unit testing. Developed changes and solutions are promoted to the CM environment, which mirrors the R19 Production environment and is used for integration level testing and troubleshooting. The SDE also serves as a staging area for changes awaiting promotion to the PSS for GAT. The code and build repositories are also co-located in the SDE.

Mobile Training Suites

The MTSs provide a stand-alone capability to deliver GCSS-MC training at the user location where network connectivity to the Enterprise Training environment is not available. MTSs consist of hardware, software, database, documentation, and procedures used in the training of GCSS-MC/LCM Increment 1 users and key personnel on the GCSS-MC/LCM Increment 1 system and processes. MTSs are built within portable enclosures so they can be easily transported, set up, and stored at any training environment. MTS cases consist of a server, storage array, 24 laptops, switches to connect the laptops to the server/array, and all ancillary equipment to have a complete computer-based classroom. As curricula are developed or updated, a Gold Disk image is sent to each major command and the image is uploaded to the MTS. MTSs are deployed to various locations worldwide.

1.3.1 System Software/Oracle Enterprise Business Suite (EBS)

In addition to EBS's out-of-the-box functionality, GCSS-MC/LCM Increment 1 has implemented Marine Corps customizations referred to as RICECPW (Reports, Interfaces, Customizations, Extensions, Configurations, Personalizations, and Workflows) adapting the EBS to meet Marine Corps specific requirements.

Table 2 provides a brief definition of the RICECPW objects. Increment 1 has implemented 250 RICE objects into the current production baseline. The system is upgrading to Release 12 (R19) and is expected to be implemented by contract award. The R19 introduces CPW (Configurations, Personalizations, and Workflows) and an extension upgrade which brings total RICECPW to 277. Additionally, the Warehouse Management System (WMS) module will introduce new RICECPW and is expected to be implemented prior to contract award.

RICECPW Object	Definition
Reports	Unique to the Marine Corps Supply and Maintenance capabilities
Interfaces	GCSS-MC/LCM connections to external systems. Increment 1 has approximately 40 interfaces
Conversions	Programs to transfer data from legacy systems to the new system. Most Increment 1 conversions are now shut off as the systems conversion is completed
Enhancements (or Extensions)	Changes to the existing EBS application that add Marine Corps functionality
Configurations	Profile options, user-defined flex fields, function and data security setups
Personalization	New prompts, titles, text, added. Existing prompt and title changes, reordering of items or regions workflows. How content is routed for review, approval, and release to the system.
Workflows	Content routing process for review, approval, and release to the system

Table 2: RICECPW Definitions

In addition to Oracle EBS, the system includes the use of other Oracle COTS products, including: Oracle Identity Management (OIM), Oracle Access Manager (OAM), Oracle Internet Directory (OID), SOA Suite, Weblogic Server, Webcache and WebGate, Mobile Field Service (MFS), Oracle Business Intelligence Enterprise Edition (OBIEE), Oracle Enterprise Manager (OEM), Oracle Business Availability Manager (BAM), Oracle Advanced Supply Chain Planning (ASCP), and Oracle Database 19c.

1.3.2 Application Implementation Methodology and Documentation

The GCSS-MC/LCM Increment 1 system architecture, design, interfaces, customizations, test scripts, operations procedures, and knowledge management tools have been developed using Oracle's Application Implementation Methodology (AIM). AIM is Oracle's project management methodology and implementation strategy. AIM provides documentation templates that support the tasks that must be performed in running and managing projects successfully.

GCSS-MC/LCM Increment 1 also adopted Oracle's Unified Method (OUM), which is a standards-based method with roots in the Unified Process. OUM is business-process and use-case driven. OUM includes support for the Unified Modeling Language (UML). GCSSMC/LCM Increment 1 Release 1.1 uses AIM. GCSSMC/LCM Increment 1 Release 1.1.1 further introduces OUM. Customizations, Personalizations, and Workflows were added to support the R19 technical upgrade.

2 GENERAL REQUIREMENTS

2.1 CONTRACT DATA REQUIREMENTS LIST (CDRL)

All CDRL deliverables shall be prepared in accordance with DoD Manual 5200.01, Volume 4, DoD Information Security Program: Controlled Unclassified Information (CUI), and submitted to the Contracting Officer's Representative (COR) and in accordance with the CDRL submission instructions.

2.2 CONTRACTOR PERSONNEL

Contractor personnel shall display badges or other visible identification that identifies them as Contractor personnel whenever they are in meetings with Government personnel. In addition, Contractor personnel shall appropriately identify themselves as Contractor employees in telephone conversations and in formal and informal written correspondence with Government personnel.

2.3 MANPOWER REPORTING

The contractor shall report contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract for the United States Marine Corps via a secure data collection site. Contracted services excluded from reporting are based on Product Service Codes (PSCs). The excluded PSCs are:

- (1) W, Lease/Rental of Equipment;
- (2) X, Lease/Rental of Facilities;
- (3) Y, Construction of Structures and Facilities;
- (4) S, Utilities ONLY;
- (5) V, Freight and Shipping ONLY.

The contractor is required to completely fill in all required data fields using the following web address:
<https://doncmra.dmdc.osd.mil>.

Reporting inputs will be for the labor executed during the period of performance during each Government fiscal year (FY), which runs October 1 through September 30. While inputs may be reported any time during the FY, all data shall be reported no later than October 31 of each calendar year. Contractors may direct questions to the help desk, linked at dod.ecmra-don-support-desk@mail.mil.

2.4 COMMON ACCESS CARD (CAC) REQUIREMENT

The COR will identify and approve those contractor employees performing on this contract that require CACs in order to perform their job function. In accordance with Headquarters, United States Marine Corps issued guidance relative to Homeland Security Presidential Directive – 12 (HSPD-12), all personnel must meet eligibility criteria to be issued a CAC. In order to meet the eligibility criteria, contractor employees requiring a CAC, must obtain and maintain a favorably adjudicated Personnel Security Investigation (PSI.) Prior to authorizing a CAC, the employee's Joint Personnel Adjudication System (JPAS) record must indicate a completed and favorably adjudicated PSI or (at a minimum) that a PSI has been submitted and accepted (opened). The minimum acceptable investigation is a T-1 or a National Agency Check with Written Inquiries (NACI). If a contractor employee's open investigation closes and is not favorably adjudicated, the CAC must be immediately retrieved and revoked.

Facility Security Officers (FSOs) are responsible for notifying the MARCORSYSCOM AC/S G-2 if any Contractor performing on this contract receives an unfavorable adjudication after being issued a CAC. The FSO must also notify the MARCORSYSCOM AC/S G-2 of any adverse/derogatory information associated with the 13

Adjudicative Guidelines/Factors concerning any contractor issued a CAC, regardless of whether a JPAS Incident Report is submitted.

Each CAC is issued with a “ctr@usmc.mil” e-mail account that the individual Contractor is responsible to keep active by logging in on a regular basis (at least twice a month), sending an e-mail and clearing any unneeded e-mails. Contractors are prohibited from “auto-forwarding” their .mil e-mail account to their .com e-mail account. If the ctr.usmc.mil e-mail account is not kept active, the G-6 will deactivate the account and CAC will lose its functionality. Contractor employees shall solely use their government furnished “ctr@usmc.mil” e-mail accounts for work supporting the USMC, conducted in fulfillment of this contract, and shall not use a contractor supplied or personal e-mail account to conduct government business. The use of a contractor or personal e-mail account for contractor business or personal use is allowed, but only when using cellular or a commercial internet service provider.

CACs will be issued to only those Contractors supporting this contract that have been authorized by the COR. The COR will authorize CACs only for those Contractors that meet current Homeland Security Presidential Directive – 12 (HSPD-12) criteria and have a definitive requirement.

If a Contractor loses their eligibility for a CAC due to an adverse adjudicative decision, they have also lost their eligibility to perform on MARCORSYSCOM contracts. CACs are not issued for convenience.

3 SPECIFIC REQUIREMENTS

Non-disclosure agreements provided by the Government are required for all contractor personnel supporting the contract. The Contractor shall coordinate all non-disclosure agreements necessary to interface with other contractors supporting this PWS in accordance with Federal Acquisition Regulation (FAR) 9.505-4. Copies of all non-disclosure agreements required for this contract shall be provided to the Contracting Officer and COR.

The Contractor shall maintain a static Internet Protocol (IP) address for individuals assigned to this contract. A static IP is required to access the tools required to conduct systems engineering, software development, code and configuration management, asset management, infrastructure management, and test activities specified in this PWS.

3.1 SERVICE MANAGEMENT

Service Level Management (SLM) provides a framework for managing Information Technology (IT) by ensuring services are defined, levels of service required to support business processes are contractual, and Service Level Agreements (SLAs) are developed and implemented to formalize and document agreements between organizations and entities. SLM processes establish clear goals for service delivery so that success factors can be established, measured, and reported. The GCSS-MC PMO has developed a metrics management plan documenting Key Performance Indicators (KPI) and metrics established for GCSS-MC/LCM Increment 1. On occasion the Operating Forces conduct User Performance Assessments (UPAs) to measure GCSS-MC/LCM Increment 1 performance against established performance objectives.

The Contractor shall establish SLM and implement a framework for managing required levels of service and service delivery goals. The Contractor shall assess current system performance and provide recommendations to enhance Quality of Service (QoS). The Contractor shall manage, monitor, and report QoS. The Contractor shall use industry best practices such as benchmarking and trend analysis to measure and enhance system performance. As new services are identified, the Contractor shall assist with defining the level of performance and capacity requirements to implement the service.

The GCSS-MC PMO has developed Level 3 process maps for fifteen (15) key sustaining engineering processes that define how GCSS-MC/LCM Increment 1 is maintained and are included as part of the solicitation's Technical Data Package. The fifteen (15) key processes are found in SDE Technical Data Package as provided in the GFI. The key processes are depicted in the GCSS-MC/LCM Program Sustaining Engineering Plan Version 1.0, Figure 6: end to end processes, page 21. Level 3 processes are documented in “Process Analysis and Narrative” documents for each

of the sustaining engineering processes. The Contractor shall coordinate and co-chair a quarterly Systems Engineering (SE) Process Integrated Product Team (IPT) and present SE process status, metrics and trends, as requested. The Contractor shall identify, communicate, discuss, plan, and execute approaches to improve SE processes and, metrics. The Contractor shall update annually the SE Sustaining Engineering Process Plan and update each process "Process Analysis and Narrative" document. For candidate processes approved by the SE Process IPT, the Contractor shall develop supporting Process Narrative documents for SE Process IPT approval.

3.1.1 System Performance and Metric Analysis

The GCSS-MC/LCM Increment 1 System Process Performance Metrics Management Plan describes the methods used to define, manage, track, and report GCSS-MC/LCM Increment 1 performance goals, requirements, and associated sustaining engineering process metrics. GCSS-MC/LCM Increment 1 metrics are used to establish a system performance baseline for measuring performance over time. Performance data is collected and evaluated to identify trends and opportunities for continual service improvement.

Within the PMO, the GCSS-MC Chief Engineer is responsible for the implementation and evaluation of the Metrics Management Plan. The PDSS lead engineer is responsible for the collection, reporting, assessment, and improvement of GCSS-MC/LCM Increment 1 performance metrics. The Contractor sustaining engineering process owners are responsible for identifying process specific metrics, collecting and analyzing them, creating process IPT presentations and materials, presenting metrics and recording minutes and action items for the SE Process IPT.

The Contractor shall evaluate GCSS-MC/LCM Increment 1 system performance requirements, and the Metrics Management Plan and measurement techniques. The Contractor shall apply IT and industry best practice and provide recommendations to enhance the effectiveness of evaluating system performance and QoS. The objective is to implement meaningful metrics that enable the PMO to make informed investment decisions to further enhance sustainment of the GCSS-MC/LCM Increment 1 system.

3.1.2 Service Agreements

To achieve defined service levels, the PMO has established several SLAs and Memoranda of Agreement (MOAs) with external service providers, including MITSC-E, MCEITS, KCITC and the Naval Systems Engineering Resource Center (NSERC). These agreements establish required service levels to enable GCSS-MC/LCM Increment 1 to achieve established service goals.

The Contractor shall review existing SLAs or propose new SLAs and provide recommendations to enhance alignment to service level objectives. The Contractor shall monitor and report on existing SLAs with external providers to ensure service providers adhere to defined service level targets. Where interdependent or internal dependencies exist for SLAs, the Contractor shall provide recommendations on establishing Operational Level Agreements (OLA) to define the required service, level of services required, and roles and responsibilities of the organizations involved. The Contractor shall identify any service level dependencies that exist in underpinning contracts (supporting contracts, such as hardware maintenance) to ensure a cohesive and integrated approach exists in executing SLM.

3.1.3 Continual Service Improvement

Continual Service Improvement (CSI) focuses on the monitoring and reporting of service and performance metrics, as well as the ongoing evaluation of the collected data to maximize the efficiency and effectiveness of the GCSS-MC/LCM Increment 1 system. Under this effort, monitoring and reporting of service and performance metrics will be required; however, no service improvement activities will be performed.

The Contractor shall monitor system performance and program processes and identify opportunities for CSI. The Contractor shall identify the approach to be used to collect supporting data, measurement and analysis, and reporting with recommendations to improve the system. ~~The Contractor shall work with the PMO to collect user surveys and provide analysis and recommendations to enhance Increment 1 performance, reduce costs, and achieve efficiencies. The Contractor shall apply cost benefit analysis in recommending service improvements to enable the Government to make informed business and investment decisions.~~

~~The Contractor shall research emerging technologies as approved by the Government and provide recommendations to the PMO.~~

3.2 SERVICE OPERATIONS

Service operations provides execution of the day-to-day processes, functions and activities necessary for delivering reliable, high quality GCSS-MC/LCM Increment 1 services. Service operations encompasses the Enterprise Service Desk, IT Operations Management, Technical Management, and Production/PSS infrastructure sustainment. GCSS-MC/LCM Increment 1 follows the Information Technology Infrastructure Library (ITIL) v3 based model for service operations. Processes managed within GCSS-MC/LCM Increment 1 service operations include request fulfillment, access management, event management, incident management, and problem management.

To execute GCSS-MC/LCM Increment 1 service operation processes and functions, the GCSS-MC PMO has established the GOC and the Enterprise Service Desk. The GOC provides for the IT, Technical, and Application Management of the Production system. The GOC is comprised of technical Subject Matter Experts (SMEs) that oversee the daily operations of the system, focusing on system operation, maintenance, application, and infrastructure sustainment. The GOC, together with the Enterprise Service Desk, provides integrated support to manage and resolve service requests, events, incidents, and problems.

The Enterprise Service Desk serves as the PMO's interface to GCSS-MC/LCM Increment 1 users. The Enterprise Service Desk provides Request Fulfillment and Access Management support to GCSS-MC users, and together with the GOC, executes Event, Incident, and Problem Management processes.

The Contractor shall execute GOC and Enterprise Service Desk services, functions and processes for the GCSS-MC/LCM Increment 1 Production system. The Contractor shall provide technical expertise in system administration, system integration, database administration, application administration, implementation of code, system software support, networking support, and Enterprise Service Desk to sustain 24-hour-a day/7-day-a-week/365-day-a-year system operations. The Contractor shall provide integrated incident/problem management to resolve issues and problems, and unplanned outages.

The Contractor shall work cohesively with the GCSS-MC Government team, including SMEs, Systems Engineering, Cybersecurity, Business System Integration (BSI), and support contractors to ensure delivery of reliable and responsive Service Operations support.

3.2.1 GCSS-MC Operations Center

The Contractor provided GOC is responsible for the operation, maintenance, and sustainment of the GCSS-MC/LCM Increment 1 Production system. Processes supported by the GOC include Event Management, Incident Management, Problem Management, Access Management, Application Management and User Community Communication. The GOC works cohesively with the Enterprise Service Desk to resolve Service Requests (SRs) to the user's satisfaction.

The GOC provides performance and event monitoring for the production system (PROD) and production support system (PSS). Event management activities include development and maintenance of event monitoring mechanisms and rules, event filtering, event correlation, and event review and closure. Event monitoring mechanisms are developed and maintained to identify and report meaningful system-generated events and set thresholds and business rules for event warnings or exception notifications. As events occur, the GOC evaluates and filters them to determine which events are informational and which events may require additional correlation and potentially require remediation activities to prevent service disruption. Events are typically captured in system logs but are not currently linked to Remedy or BMC's Information Technology Service Management (BMC ITSM). Event thresholds and notification business rules can be provided within the system logs or through external monitoring and alert systems. Each event creates an event record. Examination of multiple events, event trends and patterns may lead to incident management or to potential changes to underlying infrastructure and configuration items.

The GOC uses various tools to perform its daily system operations mission, including health checks, performance tuning, troubleshooting, and repair of system issues and faults. The GOC is comprised of personnel experienced in system, database, network and application administration, information security/cybersecurity, system integration, interfaces and data management. The GOC must also provide personnel experienced in Marine Corps supply and maintenance functions to troubleshoot and resolve system functional issues, such as transaction processing.

The GOC shall be physically located in a secure contractor facility within a 30 miles radius of Marine Corps Base-Quantico.

The Contractor shall manage and operate the GOC. The Contractor shall designate a senior technical expert as the GOC Manager. The GOC Manager shall be responsible for managing the GOC activities. The GOC Manager shall ensure proactive monitoring of the production system reports to identify events, address incidents and problems, resolve potential issues or bottlenecks, and perform corrective actions. The GOC Manager shall apprise the Government on the status of issues and provide root cause analysis with recommendations to resolve the issue. The GOC Manager shall present updates to the Government at the weekly Operations & Sustainment (O&S) meeting and prepare and deliver an O&S Status Brief to the PMO summarizing system status, performance trends, and activities.

The Contractor shall provide necessary skill sets necessary to sustain the GCSS-MC/LCM Increment 1 system. The Contractor shall perform system administration and maintenance in accordance with existing Standard Operating Procedures (SOPs) and industry best practices to ensure that system availability meets requirements and user expectations. The Contractor GOC shall monitor system health and performance in accordance with established government procedures, operations manuals and SOPs. The Contractor shall implement system performance tuning and improvement activities to ensure the most efficient use of system resources. The Contractor GOC shall perform system tuning to maintain system performance and availability. The Contractor GOC shall develop and implement system health and performance corrective action procedures as required. The Contractor shall provide recommendations for continual service improvement that enhance system performance, achieve efficiencies, or reduce operating costs.

The Contractor GOC shall identify, coordinate, and execute with the troubleshooting efforts with the KCITC in response to unplanned issues, poor network performance, hardware upgrades, Operating System (OS) upgrades, or software related issues. The Contractor shall provide recommendations on changes to the PMO SLA with KCITC that enhance production coordination and support execution between GCSS-MC/LCM Increment 1 and KCITC.

The Contractor GOC shall resolve unplanned system outages and periods of unavailability. The Contractor GOC shall also coordinate planned system outages, including providing notification to the user community before and during outages, to maximize system availability and minimize operational and user impacts.

The Contractor shall maintain, recommend updates, and implement updates to GOC related documents, including existing procedures, SOPs, and technical documents to reflect current processes, procedures, and technical implementation. All GOC technical data are expected to be Government-funded and therefore expected to come to the government with unlimited rights. Any such data that the contractor expects to restrict must be identified in a proper and timely assertion of restrictions listing. All GOC data, technical or non-technical, shall be delivered to the government, regardless of any data rights restrictions during transition at the end of the contract.

The Contractor GOC shall troubleshoot and resolve database layer, Java Virtual Machine layer, OS layer, EBS applications or Procedural Language/Structured Query Language (PL/SQL) layer and front-end/Hyper Text Transfer Protocol with Secure Socket Layer (HTTPS) layer issues.

The Contractor GOC shall manage, schedule, and coordinate both planned and unplanned outages and initiate notifications with the PMO in the event of planned emergency outages. The Contractor GOC shall initiate, request, coordinate resources, monitor, and manage all required GCSS-MC/LCM Increment 1 PROD and PSS planned and

unplanned outages required for system maintenance. The Contractor GOC shall coordinate with the PMO to develop user notifications and notify users of both planned and unplanned outages and changes to the system's functionality.

The Contractor GOC shall reconfigure Oracle Fusion middleware and Java components as required to maintain system performance and identify non-performing applications or inordinate resource requirements. The Contractor GOC shall monitor workflow error counts to ensure the system does not have a problem transacting. The Contractor GOC shall coordinate with the PMO functional/technical leads to troubleshoot and resolve workflow errors in accordance with existing SOPs, such as AIM TA.155 documentation.

The Contractor GOC shall bulk correct most workflow errors by submitting scripts to the database administrators and manually inspect and act on workflows and notifications not covered by corrective scripts. If new types of workflow errors are detected, the Contractor GOC shall investigate the cause(s) and execute corrective course(s) of action for the workflow communicating a fix or TA.155PROD24 SOP update via a Change Request (CR).

The Contractor GOC shall review, coordinate, and manage the execution of automated Build Release patches, Patch Sets, Emergency patches, and Oracle Quarterly Critical Patch Updates (CPUs) / Information Assurance Vulnerability Alerts (IAVAs). The Contractor GOC shall perform roll-back activities of Build Release patches in the event of an issue with the release. The Contractor GOC shall assist with implementing code, configuration, and design changes into the PSS. Upon completion of GAT and approval by the Government, the Contractor GOC shall deploy new software releases and patches into the Production environment.

3.2.1.1 Enterprise Application Administrator (EAA)

The EAA is responsible for application setup, configuration, maintenance, upgrades, migrations and remediation of enterprise applications. The EAA maintains system applications and ensures applications are functioning properly, backups are taken, and growth, patching, and application end-of-life are monitored and managed. As many applications are reliant on properly configured and running databases, the Contractor EAA shall work with infrastructure and development team members to ensure the proper operations of applications and databases within the GCSS-MC environments.

The Contractor EAA shall resolve application problems not resolved at the Tier 3 level. Support may include coordinating vendor and host site support. The Contractor EAA shall collaborate with and forward problems to the appropriate entity for resolution and corrective action.

The Contractor EAA shall analyze and provide support for workflow processes and problems, and monitor system integrity and usability in accordance with the current metrics requirements. The Contractor shall perform Root Cause Analysis (RCA) for Priority 1 (P1) and Priority 2 (P2) issues and for problem management issues (CDRL A001).

The Contractor shall manage and maintain a Knowledge Base (KB) (CDRL A002) to facilitate efficient and effective resolution of user problems (currently maintained in Remedy ITSM). New KB articles, templates, and procedures recommended by the EAA shall be submitted for approval by the PMO (CDRL A002) via the GCSS-MC Training Resource Manager (TRM) and the Incident Management and Problem Management workflow processes. The Contractor shall develop and submit CRs that require Oracle COTS changes to the GCSS-MC GOC Manager for escalation to the Configuration Control Board (CCB).

The Contractor shall conduct technical aspects of functional user testing and collaborate with the PMO on code fix designs in accordance with the Government's Configuration and Change Management processes.

Deliverable

CDRL A001 – Scientific and Technical Report - (RCA Report)

CDRL A002 - Scientific and Technical Report - (Knowledge Base (KB))

3.2.1.2 System Administration

The Contractor shall provide 24-hour-a day, 7-day-a-week, 365-day-a-year system administration (SA) event and performance monitoring for application, Web tier, OS and below, servers, hardware, third-party software, storage, network appliances, Riverbed components, f5 load balancer, and administrative user maintenance used within the Production system. SA shall manage the production hardware warranties, resolve SRs, and perform system backups. SA shall perform necessary server maintenance, research hardware, operating system issues, and apply patches and updates to Exadata, Oracle Virtual Machine, Oracle Virtual Server, Infiniband, Network Switches, Management Switches, Management Server, and Virtual Machines. SA shall support IAVA management to ensure the system remains compliant to IA/Cybersecurity certification requirements.

3.2.1.3 Database Administration

The Contractor shall apply IT industry best practices in database administration to maintain and sustain the production databases. The Contractor Database Administrator (DBA) shall perform health checks and monitor system integrity and usability in accordance with the current requirements, AIM TA.155, or operating procedures. The Contractor DBA shall monitor database performance, maintain database integrity, apply necessary patches, and perform preventive and corrective actions to sustain and improve system performance.

The Contractor DBA shall use existing Oracle monitoring tools to diagnose database issues and conduct performance tuning and optimization. The Contractor DBA shall administer space usage for table spaces within the environments and provide performance tuning and maintenance of the environments.

The Contractor DBA shall identify system event thresholds and implement proactive response alerts, respond to alerts, and initiate the incident management process pursuant to service restoration. The Contractor DBA shall resolve data correction issues identified through monitoring and SR submissions.

The Contractor DBA shall provide the government functional design presentations, functional application testing, functional impact assessments, and recommendations for business procedure updates as required.

The Contractor DBA shall grant system access privileges as specified by GCSS-MC Information System Security Manager (ISSM) based on pre-defined roles (which will then be granted to database users) or directly to user accounts.

The Contractor DBA shall perform weekly and on-demand backups and initiate recoveries and refreshes when required for duration of the contract.

The Contractor DBA shall monitor the growth of the database segments and conduct capacity planning and forecasting. The Contractor shall report and present this data to the Government at the Weekly O&S meeting.

The Contractor DBA shall create and maintain database links to the environments and perform table and/or table space defragmentation operations to address any fragmentation issues, as necessary.

The Contractor DBA shall implement any product upgrades, such as Service Oriented Architecture/ Business Process Execution Language (SOA/BPEL), Grid Control, OBIEE, B2B migration, etc. as approved by change management.

The Contractor DBA shall monitor production interfaces to support problem identification and create CRs to address valid issues.

The Contractor DBA shall analyze issues forwarded to Tier 3 by Tier 2 (CDRL A001). The Contractor DBA shall research, design, build, test, and promote approved changes per the Government's Service Transition processes, monitor changes to RICECPW, and validate patches to RICECPW components.

The Contractor DBA shall apply build/emergency/patch sets to Production environments and apply quarterly security patches as they become available by Release Management.

The Contractor DBA shall manage database environments to support troubleshooting production support problems, applying patches, and testing.

The Contractor DBA shall analyze scope (CDRL A001), prepare, and execute CRs which may arise from high priority CRs authorized by the change management process and recommend via the change management process those changes that may provide more effective use or reduce maintenance costs for the GCSS-MC/LCM Increment 1 system, from what is in the currently implemented business rules.

The Contractor shall maintain Oracle wallets and Public Key Infrastructure (PKI) certificates for Oracle applications.

Deliverables

CDRL A001 – Scientific and Technical Report - (RCA Report)

3.2.1.4 Systems Integration

The Contractor shall integrate all hardware, software, applications, data, network, and component subsystems into an efficient single system. The Contractor shall provide daily workflow administration and troubleshooting to prevent or resolve redline and document script issues and provide RCA and resolution recommendations for issues requiring additional resources (CDRL A001) or change management approval.

Deliverable

CDRL A001 – Scientific and Technical Report - (RCA Report)

3.2.1.5 Interface Administration

The Contractor team shall coordinate with GCSS-MC/LCM Increment 1 external partners to maintain efficient interfaces required to satisfy user data requirements. Such coordination may take the form of code promotion, data or process changes, infrastructure upgrades, new subscriber support, external partner coordination meetings (IPTs, Working Groups), or other necessary support the interface capability groupings listed in Table 3:

Interface Capability Grouping
Master Data Management (Referential Base Files)
Force Structure & Personnel Data
Customer and Vendor Data
User Registration
MRP, DoD Source Supply, CMOS, GCSS Army, GSA
Legacy Systems Warehousing
Core \$\$\$ (funding transactions)
RF-ITV (Radio Frequency In-Transit Visibility)
War Reserve System
Depot

Table 3: GCSS-MC/LCM Increment 1 System Interface Capability Groupings

The Contractor shall analyze, troubleshoot, and resolve interface issues between GCSS-MC/LCM Increment 1 and external systems. The groupings of interface capabilities are current as of the time of the publishing of the

solicitation; however, the groupings and the individual interfaces that make them up are subject to change by the time of contract award.

The Contractor shall monitor interfaces and make necessary immediate, responsive configuration changes to system interfaces to support operational needs. The Contractor shall properly document immediate and formal CM/CCB approved changes in appropriate Interface Control Documents (ICDs), System Interface Agreements (SIAs), and technical drawings (CDRL D001). The Contractor shall work closely with the GCSS-MC Cybersecurity team to coordinate updates to the Ports, Protocols and Services Management (PPSM) and maintain system accreditation. The Contractor shall provide daily, weekly, and periodic Interface and External Tracking Reports with required interface metrics interface status during the past 24 hours, including interface status, downtime over the total records, total success, and success percentage. The Contractor shall analyze tracking reports and provide holistic assessments on GCSS-MC/LCM Increment 1 system impacts and provide recommendations for solutions.

The Contractor shall provide daily, weekly, and periodic Error Handling Framework (EHF) reports for RICECPW. The Contractor shall analyze reports and provide holistic assessments on GCSS-MC/LCM Increment 1 system EHF impacts and provide recommendations for solutions.

The Contractor shall provide technical assistance to Service Desk tickets, Incident and Problem Management to quickly identify, assess, and resolve changes and system outages. The Contractor shall provide a rapid response capability, conduct root cause analysis, perform and document necessary changes to interfaces and external reports to support operational requirements. The Contractor shall coordinate with internal and external interface stakeholders to troubleshoot interface issues, determine corrective action, and implement approved solutions.

The Contractor shall maintain awareness of production build impacts to workflows to determine if the fix worked and/or created additional errors.

The Contractor shall bulk correct most workflow errors by submitting approved builds/scripts to the Contractor DBAs.

The Contractor shall implement approved new or revised Increment 1 interfaces.

Deliverable

CDRL D001 - Technical Report - Study/Services - (Technical Data Package)

3.2.1.6 Data Quality

GCSS-MC/LCM Increment 1 processes more than 12–15 million records and 3 million transactions per day. The Contractor shall provide day-to-day data management to support the end-to-end process for managing the lifecycle development and collaboration points between the COTS software, system customizations, external systems, and third-party software.

The Contractor shall maintain data quality to ensure efficient business and transaction processing functions. The Contractor shall apply in-depth knowledge of data, data flow, and data structure to analyze issues, provide recommendations, and implement approved solutions to resolve issues through performance monitoring, incident and problem management. For issues involving records that are stuck in processing, or need to be reprocessed, the Contractor shall develop and execute approved scripts to appropriately clear the issue. If the script requires repeated execution the contractor must create a CR to modify the TA.155PROD 24 to run the program concurrently.

The Contractor shall conduct end-to-end analysis of data quality, integrity, and availability of GCSS-MC/LCM Increment 1 data, integrating both functional and technical perspectives and perform analysis of system changes resulting from data-related governance directives and policies.

The Contractor shall conduct data cleansing and reporting activities to system owners necessary to identify and correct data inconsistencies, such as incorrect formats, invalid values, or duplicate records.

The Contractor shall collect, compile, and evaluate GCSS-MC/LCM Increment 1 enterprise data discrepancies and conduct technical review of data from collection to instantiation (data lifecycle) within the system. The Contractor shall report and present this data to the Government at the Weekly O&S meeting.

The Contractor shall support the technical/functional integration of GCSS-MC/LCM Increment 1 data, maintenance of data quality and integrity, and analysis of impacts to GCSS-MC/LCM Increment 1 for all system changes (planned and unplanned).

The Contractor shall identify and analyze trends and submit recommended CRs for enhancements that improve data quality.

3.2.1.7 Information Assurance/Cybersecurity

The Contractor shall provide ITSM ticket assessments to assist the GCSS-MC/LCM Increment 1 Cybersecurity team with user accounts and system access issues and adjudicate at the lowest level or reassign tickets to the GCSS-MC Cybersecurity team.

The Contractor shall support Role Based Access Control (RBAC) implementation by maintaining components, including OIM, OID, and EBS.

The Contractor shall support system use of Electronic Data Interchange Personal Identifier (EDIPI) and provide Oracle Web certificate wallet management.

The Contractor shall conduct user account auditing.

The Contractor shall conduct audit log reviews and provide anomalies to the Program ISSM.

The Contractor shall provide the Program ISSM with augmented GOC support to maintain the Authentication Stack base software and RICECPW.

3.2.2 Enterprise Service Desk

The GCSS-MC Enterprise Service Desk is the initial customer point of contact for worldwide GCSS-MC/LCM Increment 1 user support. The Service Desk responds to a variety of SRs, including account and access requests, technical SRs, requests for change, and service issues. The Enterprise Service Desk responsibilities include Request Fulfillment, Access Management, Event Management, Incident Management, and Problem Management. On average, the Service Desk processes more than 120 SRs daily.

The overarching objective for the Service Desk is to provide high quality support in a timely and responsive manner and achieve a high level of customer satisfaction. A further objective is to implement a 'Tier 0' capability that provides users with self-help tools, processes, and information to support user needs, thereby reducing the number of SRs and the demand on Service Desk resources. New KB articles, templates, and procedures recommended by the Service Desk shall be submitted to the PMO for approval (CDRL A002).

GCSS-MC/LCM Increment 1 currently uses the BMC ITSM tool to track and manage SRs. The PMO maintains a historical record of SRs and will provide these to the Contractor as Government Furnished Information (GFI).

The Service Desk is responsible for maintaining and using the Service Desk Knowledge Management Repository. The repository contains KBs that document pre-determined procedures to assist Service Desk personnel and GCSSMC/LCM Increment 1 users with resolving SRs. The repository will be provided to the Contractor as GFI. In addition to the KBs, the Service Desk has access to User Productivity Kits (UPKs). NOTE: Oracle may no longer support the software used to create UPKs after 2022 and the name of the new software is not yet known. UPKs are developed training artifacts that provide GCSS-MC/LCM Increment 1 users with detailed instructions on how to use the system. The Service Desk provides users with links to the UPKs that provide information to satisfy the service

request. The Service Desk also uses existing classification guidance for issues or defects (referred to as system, component, item, module (SCIM)) to properly identify and route the issue or defect.

The Contractor shall establish and manage an integrated Enterprise Service Desk. The Contractor shall apply industry best practices to manage Service Desk operations throughout the lifecycle of SRs, including establishing, routing, tracking, closing, and reporting of all Service Desk requests. The Contractor shall apply industry standard processes such as ITILv3 or Capability Maturity Model Integration (CMMI) Level 3 processes to manage and execute the Service Desk and provide continual service improvement to enhance service and reduce costs.

The Contractor Service Desk shall accomplish support at the lowest Tier level. When necessary, the Service Desk shall escalate and track requests until resolved to the customer's satisfaction. The Contractor shall ensure Service Desk personnel are properly trained in GCSS-MC processes and remain current with GCSS-MC/LCM Increment 1 system functions, processes, workflows, and applications. The Service Desk shall manage all web, email, phone, voicemail, walk-in, and ticketing system submitted customer inquiries from receipt to resolution using established tools, including BMC Remedy. The Contractor shall work with the PMO and with the Logistics Systems Coordination Offices (LSCO) to identify approaches that enhance Tier 0 Self-Help and enhance LSCO support at the local level.

The Contractor shall use IT industry best practices to identify and use relevant Service Desk data. The Contractor shall work closely with the Government to identify relevant data, perform trend analysis and provide Service Desk Activity Reports that analyze SR data and trends at the Government's weekly O&S meeting. The Contractor shall identify issues or anomalies, resolve issues, and provide continual service improvement recommendations to enhance cost-effective Service Desk support.

The Contractor shall collect and report customer satisfaction surveys to ensure Service Desk delivery meets customer expectations for quality and timeliness of support.

Along with the GOC Manager, the Service Desk lead shall report the Service Desk's efforts and status at the Government's weekly O&S meeting.

Deliverable

CDRL A002 - Scientific and Technical Report - (Knowledge Base (KB))

3.2.2.1 Tier 1 Service Desk

The Contractor shall provide a 24-hour-a-day, 7-day-a-week, 365-day-a-year fully functional Tier 1 Service Desk. All personnel assigned to Service Desk support must speak fluent English as calls may come in from Operating Forces with degraded communication circuits.

The Service Desk shall identify and differentiate between technical and knowledge issues. When addressing technical issues or defects, the Tier 1 Service Desk shall use existing classification of the issue or defect (referred to as SCIM) to properly identify and route the issue or defect. The Tier 1 Service Desk shall facilitate the flow of information between users and internal Government and Contractor teams, including SMEs, IA, BSI, and the GOC. For knowledge issues, the Tier 1 Service Desk shall use established UPKs to satisfy service requests, including system access issues and basic system functions. If a UPK does not exist, the Contractor shall create a new KB (CDRL A002) or UPK to document the solution and submit it to the Government for review and approval. The Contractor shall assist the PMO in identifying trends in inadequate end-user training to enhance GCSS-MC/LCM Increment 1 training curricula in response to end-user needs. When applicable, the Tier 1 Service Desk shall create a new KB when a reoccurring SR issue response is not already scripted in Remedy. The Tier 1 Service Desk shall ensure all requests have been properly logged into the Service Desk system and resolved to the user's satisfaction before closing a Service Desk ticket. Service Desk tickets are NOT closed when elevating the issue to the next level.

The Tier 1 Service Desk shall report resolution of issues and trouble tickets/SRs to the user via the established Frequently Asked Questions (FAQ) site within BMC ITSM. The Contractor shall maintain and further develop the GCSS-MC Tier 0 Self-Help approach for customer assistance including GCSS-MC/LCM Increment 1 UPKs and the public facing KB and FAQ website.

The Contractor shall coordinate with the Government and perform a review cycle of Service Desk artifacts to ensure accuracy and relevance of the artifacts. The Contractor shall identify which artifacts are useful to the user and provide recommendations on enhancing direct access to them, including KBs and FAQs.

Deliverable

CDRL A002 - Scientific and Technical Report - (Knowledge Base (KB))

3.2.2.2 Tier 2 Service Desk

The Contractor shall provide a Tier 2 Service Desk. The Tier 2 shall provide knowledge on the use and navigation of the standard features and functionality for the applications established for GCSS-MC/LCM business processes. The Tier 2 Service Desk shall work closely with Tier 1 to ensure requests have been resolved to the user's satisfaction and that the resolution is fully documented in BMC ITSM. The Contractor shall update the GCSS-MC/LCM KB (CDRL A002) when a Tier 2 level SR issue response is not already scripted or documented in the KB for reoccurring issues. The Tier 2 Service Desk shall analyze issues to determine if possible trends exist and report them at the Government's weekly O&S meeting. Requests that cannot be resolved at Tier 2 shall be elevated to Tier 3 in accordance with established criteria. The Tier 2 will assess the Service Request, determine corrective action, and coordinate with the Tier 3 for execution of corrective action.

Deliverable

CDRL A002 - Scientific and Technical Report - (Knowledge Base (KB))

3.2.2.3 Tier 3 Service Desk

The Tier 3 Service Desk receives SRs that may not be resolved at the Tier 1 or Tier 2 Service Desks and provides functional, technical support, and consultation services to the PMO. The Contractor shall provide SMEs for troubleshooting, performing RCA, solution development and implementation of approved changes in order to resolve SRs, data quality, integration, data exchanges, and customization. SR solution development and implementation shall be in accordance with the Government's Service Transition and Systems Engineering processes and deliverables.

The Contractor shall provide a Tier 3 Service Desk comprised of functional and technical SMEs to provide timely and effective responses to operational or technical problems or issues that may arise during system operation and maintenance. The functional SMEs shall possess expertise in Marine Corps supply and maintenance functions and processes. The technical SMEs shall possess expertise in data and data flow analysis, database and system administration, system integration, Oracle EBS application, and cybersecurity. When required, the Tier 3 support team shall work cohesively with the Government's SME team, the other components of the GOC, LSCO, and Deputy Commandant, Installations and Logistics (DC I&L) Liaison Officers to identify and resolve issues, incidents, and problems through the GCSS-MC PMO.

The Contractor shall provide on-call and extended shift Tier 3 support. In the event of a problem, application, or system outage requiring immediate attention, Tier 3 support shall be available within a two (2) hour window to perform and complete required tasks to resolve the issue. Tier 3 personnel shall also be available to work extended hours to perform mission essential tasks to include Host Site interface or coordination, capability fielding, and cutovers.

3.2.2.4 Service Desk Metrics

Performance metrics for the service desks are captured in Table 4 below with the respective mechanism used for the information to be reported.

Key Metrics	Threshold	Objective	Reporting Mechanism
First Contact Resolution (FCR)	60%	70%	BMC
Tier 1 Resolution Rate	50%	70%	BMC
Tier 2 Resolution Rate	80%	90%	BMC
Abandon Before Answer Rate	< 3%	< 1%	Avaya
Average Speed to Answer (ASA)	<20 seconds	<10 seconds	Avaya
Average Speed to Answer Email/Web	< 1 hour	< 30 minutes	BMC
Customer Satisfaction	90%	100%	BMC
Average Time to Resolve (TTR) – Tier 1	< 10 hours	< 5 hours	BMC
Average Time to Resolve – Tier 2	<36 hours	< 16 hours	BMC

Table 4: Service Desk Metrics

All incoming incidents are used in determining resolution and abandon before answer rates. FCR, Tier 1 and Tier 2 resolution rate excludes known Tier 3 issues at time of receipt (account reactivation, cybersecurity, etc.). Average TTR is based on the time from first contact of the issue. TTR-Tier 1 is based on a 24-hour/7-day a week clock. TTR-Tier 2 is based on an 12-hour workday and 5-day a week clock.

3.2.3 Incident Management (IM)

The primary focus of IM is the restoration of services following an incident. IM primarily reacts to provide guidance on diagnostic and escalation procedures required to make quick restoration of services.

The Contractor shall provide assessment, diagnostic, and escalation procedures to resolve incidents within specified timeframes and ensure restoration of services in accordance with GCSS-MC/LCM Increment 1 availability requirements. The Contractor shall gather information regarding the incident and provide incident classification based on the SCIM of identified incidents and provide trend analysis at the Government's weekly O&S meeting. The Contractor shall record incident details in the Service Desk Ticketing System and prioritize incidents in terms of impact and urgency with the objective to minimize user impact. Upon resolution of the incident, the Service Desk shall verify and document in BMC ITSM the incident has been resolved to the satisfaction of the affected user(s) before proceeding with closing of the SR. If incidents cannot be resolved technically or in a timely manner, the Contractor shall escalate the issue to Problem Management for further analysis and resolution.

3.2.4 Problem Management

The Contractor shall identify and apply the necessary skills to troubleshoot problems escalated to Tier 3. The Contractor shall provide corrective maintenance to resolve all level trouble tickets/SRs and notifications. The Contractor shall research technical issues identified through the Incident and Problem Management processes and perform support for data quality, integration, data exchanges, and customizations problems. The Contractor shall identify and resolve report and attempt to facilitate the resolution of data integrity issues and perform analysis of problems related to data contained in the Oracle EBS applications and GCSS-MC/LCM Increment 1 environments. The Contractor shall conduct analysis (CDRL A001) of workflow process incidents and problems within GCSS-MC/LCM, troubleshoot data integrity issues within the GCSS-MC/LCM Increment 1 system, and support troubleshooting efforts with external interface partners.

The Contractor shall analyze and resolve functional issues and provide technical and functional support for the GCSS-MC/LCM Increment 1 system to include COTS products, RICECPW components, and related trouble tickets/SRs that could not be resolved at lower tiers. The Contractor shall identify root causes, perform RCA, and document their findings in an RCA report provided (CDRL A001) to the Government. The Contractor shall manage Tier 3 support, coordinating with external interface partners and with application developers to obtain and verify system fixes. The Contractor shall coordinate vendor support including Oracle product support to resolve identified incidents and problems. The Contractor shall obtain and validate data fixes if the root cause of the problem is the Oracle COTS product and provide to the PMO for inclusion in CCB requirements determination.

The Contractor shall address problems logged via the IM process in accordance with the Government's Problem Management Process and shall record problem details, to include RCAs, workarounds, and resolutions, in the Problem Record within the Service Desk Ticketing System. Some problem solutions may require escalation to the Government's Service Transition and Systems Engineering processes.

The Contractor shall assist with workarounds and non-software update activities to address resolution of problems related to data and provide recommended workarounds/problem resolution to address the root cause and reduce recurrence of repetitive problems.

The Contractor shall use historical data to support predictive analysis to eliminate potential incidents and problems before they occur and to identify workarounds they shall report at the Government's weekly O&S meeting.

The Contractor shall conduct a technical review of system data and provide ad hoc reports or data extracts as it relates to USMC logistics business processes.

The Contractor shall maintain and implement SOPs to address GCSS-MC/LCM Increment 1 user needs and provide information to address specific needs/issues forwarded from Headquarters (HQMC) I&L and the GCSS-MC/LCM Marine Corps Logistics Command (MARCORLOGCOM) user representative.

The Contractor shall analyze reported issues (CDRL A001) and identify potential negative trends including those mitigated by training, workarounds, or changes to AIM TA.155 SOPs.

The Contractor shall maintain records of unresolved issues and report those issues in accordance with the current escalation procedures. The Contractor shall identify, develop, and submit CRs to Change Management to document known problems that may require further research or long-term resolution.

Deliverable

CDRL A001 – Scientific and Technical Report - (RCA Report)

3.3 GCSS-MC PRODUCTION/PSS ENVIRONMENT SUSTAINMENT

The Contractor shall maintain and sustain the current production/production support system in the GCSS-MC Cloud One environment. The Contractor shall support troubleshooting and corrective action in response to events, incidents, or problems. The Contractor shall maintain configuration control of the system baseline and ensure all changes to production configuration items are properly documented and reported. The Contractor shall maintain control of production assets and support physical inventories and configuration audits. The Contractor shall apply security patching and risk management framework (RMF) artifact updates to ensure system accreditation and sustainment of its Authority to Operate (ATO). The Contractor shall resolve performance and functional anomalies.

The Contractor shall cohesively integrate and sustain the Production environment and ensure its synchronization with the SDE and training environments.

The Contractor shall provide recommendations for improvement for potential software and proposed hardware used in the Production environment via ECPs utilizing the Engineering Change Proposal.

3.3.1 Capacity Management

The Contractor shall manage current IT resources and ensure resources are effectively used to meet current capacity requirements. ~~The Contractor shall identify current application and infrastructure bottlenecks for corrective action, system utilization, growth patterns, and future demands that impact system capacity so proactive actions can be taken by the PMO to adjust system resources.~~

The Contractor shall perform capacity planning to accommodate for usage growth through scaling of existing infrastructure (servers, licenses, storage, bandwidth, etc.). This growth can include new users, higher levels of

usage (more concurrent users), hosting and integration of additional applications and web services, continued migration of capabilities to the GCSS-MC/LCM Increment 1 system, connection of additional interfaces or data feeds, creation of new dashboard/analytics applications and services, and federation with other systems.

The Contractor shall provide a weekly GCSS-MC Enterprise NIPRNet Capacity Report that includes information related to service, resource use, and performance. The Contractor shall document current system use and forecast requirements, to maintain an annual Capacity Plan. The Contractor shall include recommended applicable machine modifications and/or additional equipment to increase the capacity of the system in the plan. The Contractor shall perform capacity and forecasting for Production based on current needs and future development plans.

3.3.2 IaaS Sustainment and System Management

The Contractor shall provide all services and support necessary to sustain the GCSS-MC system, which is fully hosted within the Government's Cloud One IaaS environment.

The Contractor's responsibilities under this section are strictly limited to sustainment-in-place and maintaining the established system baseline. No new work, capability enhancements, or feature developments shall be performed. The only exceptions are work required to resolve a system failure (break/fix) or to remediate a cybersecurity vulnerability.

Cloud Environment Governance: The Contractor shall act as the technical liaison between the GCSS-MC PMO and the Cloud One hosting provider. The Contractor shall:

- Monitor and report on the provider's adherence to established Service Level Agreements (SLAs).
- Advise the PMO on technology changes only when required to maintain security, ensure continued supportability, or address component obsolescence.
- Provide analysis and recommendations on Cloud Service Provider CRs to assess their impact on the existing GCSS-MC baseline.

System, OS, and Database Administration: The Contractor shall perform all administration required to maintain the health and performance of the system's as-is configuration. The Contractor shall:

- Conduct system administration of the Red Hat Enterprise Linux and Windows Operating Systems, limited to break/fix, security patching, and issue resolution.
- Perform all Oracle database, Oracle E-Business Suite (EBS), and middleware monitoring, performance tuning, and patching. Installation or re-installation of software components is authorized only as required for break/fix activities or vulnerability remediation.

Security and Continuity of Operations: The Contractor shall ensure the system maintains its security posture and is prepared for contingency operations. These activities are considered mandatory sustainment. The Contractor shall:

- Apply all required security patches, including Oracle quarterly Critical Patch Updates (CPUs), and resolve all directed Information Assurance Vulnerability Alerts (IAVAs).
- Ensure the system maintains compliance with all applicable Security Technical Implementation Guides (STIGs).
- Manage and test the system's established backup and recovery solution.

Configuration and Interface Management: The Contractor shall maintain accurate documentation of the current baseline and manage existing interfaces. The Contractor shall:

- Maintain and provide accurate system documentation reflecting the as-is architecture, including system, network, and data flow diagrams.
- Manage existing GCSS-MC user accounts and system permissions.
- Coordinate troubleshooting and testing for existing system-to-system connections and data exchanges. No new interfaces shall be developed.

Strategic Support: The Contractor's role in strategic activities is strictly limited to providing analysis and information about the current system.

- If directed by the Government, the Contractor shall provide technical data and analysis to support planning for the eventual decommissioning of the IaaS environment. This support does not include the execution of any migration or transfer of services.

Deliverable

CDRL D001 - Technical Report - Study/Services - (Technical Data Package)

3.3.3 Internal Networking and Ancillary Storage

The Contractor shall ensure the delivery of secure, reliable, and scalable network connections, to include specifications for establishing a link between Contractor sites, Contractor personnel, and Government sites. The Contractor shall be responsible for access control into contractor environments and to the Marine Corps environment. The Government and Contractor are responsible for access control within their respective networks.

The contractor shall prepare CRs to address COTS updates and configurations changes for the f5 and network switches.

The Contractor shall monitor and tune multi-protocol support of NetApp devices, monitor system logs, and configure approved settings of network switches. The Contractor shall troubleshoot and update settings for Network Information Service (NIS), Domain Name Service (DNS), Internet Message Access Protocol (IMAP), Simple Mail Transfer Protocol (SMTP), and Dynamic Host Configuration Protocol (DHCP) servers. The Contractor shall provide technical support in configuring the SDE to replicate the Production environment's configuration.

3.4 RESERVED

3.5 ENTERPRISE TRAINING AND TRAINING DEVICES

The Contractor shall provide an integrated approach to training management, training change management, delivery and training device sustainment ensuring GCSS-MC/LCM Increment 1 training and training devices mirror the functions of the Production system.

3.5.1 Training Studies

The Contractor shall conduct training studies to assess and recommend new training methodologies and approaches, and feasibility studies for training development, technical support, and training documentation when requested to by the Government.

3.5.2 Training Management

The Contractor shall ensure training materials, training curriculum, training delivery, and training systems accurately reflect the capabilities, functions, and processes of the GCSS-MC/LCM Increment 1 system. The Contractor shall manage the training curriculum transition to formal schools throughout the Marine Corps Training and Education Command (TECOM) and MARFORs via the "Gold Disk." If this process changes due to the potential replacement of the MTS, the Contractor shall become familiar with the new processes used to properly manage the training curriculum transition to the formal schools throughout TECOM.

The Contractor shall designate a Training Resource Manager (TRM) who will manage the training program during all phases of training curricula development and delivery. The TRM shall evaluate changes to the GCSSMC/LCM Increment 1 system and advise the PMO on impacts to training materials, curriculum, training, and training systems. The Contractor shall ensure all training courses are kept current. The use of errata or interim changes is to be limited. No more than two (2) errata may be issued for any single course within one (1) fiscal year. If a course requires more than two errata in a fiscal year, the course must be formally updated to incorporate all changes into the

core curriculum. All courses are considered due for update and must be maintained to reflect the current system baseline. The TRM shall provide a Training Schedule with Plan of Action and Milestones (POAM) for all training changes and updates to ensure training curriculum, materials, training and systems are updated in a timely manner in accordance with approved standards. The Contractor shall provide Status Reports of ongoing training activities in support of change management or other PMO processes. These reports shall include a quarterly training matrix with the data points specified in the table below.

Quarterly Training Delivery Matrix

Course Name	Duration (Hours)	Training Type (In-Person/Hybrid)	Iterations This Quarter	# of Hands-on Trainings per Iteration	Attendee Rank	Attendee Role	Attendee Unit Location	Course Location

3.5.3 Training Development

The Contractor shall provide personnel with Oracle applications DBA and EBS experience to develop and deploy training courseware and materials to GCSS-MC training locations. The GCSS-MC/LCM Increment 1 Enterprise Training encompasses training materials housed at the Marine Corps TECOM FLCs, the Marine Expeditionary Force (MEF) Materiel Readiness Centers (MRTC), and the DISA Global Content Delivery System (GCDS). The FLC training materials consist of supply, maintenance, and finance military occupational specialty entry-level and career-level courseware. The MRTC training materials support MEF supply, maintenance, and financial supply and maintenance sustainment training activities. Currently, the MTS is the designated GCSS-MC/LCM Increment 1 training device. In the event the MTS is replaced with a new training device, the Contractor shall become familiar with the new GCSS-MC/LCM training device.

The Contractor shall follow all CM and release management requirements for training material and systems. This includes managing training material items as configuration items, ensuring creation of end-user interim training materials, incorporation into courseware, and retirement of outdated items. The Contractor shall maintain the versioning, consistency, and accuracy of training courseware across the training environments, including MarineNet, DISA GCDS, and the MTS. The Contractor shall develop and update training materials in accordance with the GCSS-MC/LCM Configuration Management Plan (CMP) and requirements predicated by the other governing organizations, including MarineNet and DISA/GCDS as well as the following references:

- NAVMC 1533.1, "Systems Approach to Training (SAT) Manual"
- MCO 1553.2B, "Management of Marine Corps Formal Schools and Training Detachments"
- MCO 1553.3A, "Unit Training Management"
- USMC TECOM College of Distance Education and Training, "MarineNet Courseware Development Technical Standards," Version 2.0 dtd July 18, 2012
- DODI 1322.26, Sharable Content Object Reference Model (SCORM)
- GCSS-MC eLearning Manual and Style Guide
- GCSS-MC Design_Standards_Release_7_V2.pdf

The Contractor shall include training materials and systems as part of the Technical Data Package (TDP) (CDRL D001).

The Contractor shall conduct user training needs analysis and associated tasks for approved system changes, including functional and technical modifications, system test events, and operations and sustainment. The Contractor shall recommend effective training methods and training delivery to address system changes. The Contractor shall evaluate proposed change(s)/update(s) to ensure the solution adheres to DoD/Department of the Navy (DON) human engineering criteria for design and development and ensure Human Systems Integration (HSI) impacts are identified and mitigated.

The Contractor shall analyze KBs to determine potential training gaps and recommend updates to ensure KBs are developed in accordance with approved training curriculum. The Contractor shall develop the training curriculum,

including Courses of Instruction (COI) and lesson plans using established design templates and Government-approved media and delivery methods. The Contractor shall identify, revise, and deliver training material updates (CDRL F001) based on software patch set updates to ensure all training materials are aligned to the current production system baseline. This includes Instructor/UPK products, student guides, instructor guides, and job aids.

The Contractor shall ensure all training materials are in compliance with all Marine Corps and Navy training standards and policies. The Contractor shall update the GCSS-MC/LCM Increment 1 training curriculum in support of formal school use and conversion of GCSS-MC training materials using the approved Marine Corps curriculum standards: the Systems Approach to Training (SAT) methodology and the SCORM format.

The Contractor shall use PMO-approved storyboards and design templates and ensure all delivered training media is based on approved scenarios using a variety of content and delivery methods, including text, still graphics, movies, animations, music, voice-over narrations, databases, Oracle UPKs, Shockwave content, PowerPoint, and Adobe products such as Captivate. The Contractor shall provide the Government full access to review and test training materials prior to acceptance. NOTE: Oracle support for UPK is scheduled to end in April 2019. A new software will replace UPK but that replacement has not yet been determined. Once the new software is named, existing UPKs will need to be upgraded using the new software whenever a change is required to the existing UPK training packet.

Upon approval and acceptance by the Government, the Contractor shall deliver a quarterly update of curriculum material, eDesktop, and associated documentation for publication to the appropriate environment, including MarineNet, DISA/GCDS, MTS, and any new training environment the Government identifies. This material includes UPKs, videos, and quick reference guides. The Contractor shall provide MarineNet and MTS deliverables in the proper format and upload materials to the DISA/GCDS. Additionally, the Contractor shall be prepared to provide the training device deliverables in the proper format and upload materials to the DISA/GCDS, if required. The Contractor shall maintain access and control of the training media library.

Deliverables

CDRL D001 - Technical Report - Study/Services - (Technical Data Package)

CDRL F001 - Instructional Media Package – (Training Material Update)

3.5.4 Training Delivery

The Contractor shall provide instructors with knowledge of the GCSSMC/LCM Increment 1 system and training experience in the Customer Relationship Management, Supply Chain Management, Finance, and Planning areas. The TRM shall coordinate with the government Manpower, Personnel and Training (MPT) lead on all training execution events. The TRM shall ensure that platform instructors have experience following the SAT teaching GCSS-MC/LCM through two or more training methods. The Platform Instructor shall coordinate with the TRM to conduct onsite training.

The Contractor shall coordinate with the GCSS-MC training led to coordinate training events and ensure the training environment is suitable and sufficient to conduct training. On average, the current local training activities consist of three consecutive days: one (1) day for setup, two (2) days for instructions and breakdown of MTS on the last day of training. The frequency of training will be at a minimum of one (1) training class per quarter with a surge capability of three (3) classes per quarter. Each class will consist of up to twenty (20) students. The Contractor is responsible for setup, configuring, operational checks and updates as necessary. The Contractor is also responsible for the loading of the training material to support training.

The Contractor shall conduct on-site comprehensive training classes that are instructor-led and, in a Government, approved platform instruction. Qualified instructors will be required to travel to CONUS/OCNUS locations. Anticipated travel is to one or more locations per quarter, including, but not limited to, I MEF (Camp Pendleton, CA), II MEF (Camp Lejeune, NC), III MEF (Marine Corps Base Hawaii and Okinawa, Japan), MARCORLOGCOM (Albany, GA), and Marine Corps Systems Command (MCSC) (Quantico, VA). The

Contractor shall also provide Government approved distant learning/web environment comprehensive training classes to supplement on-site training.

The Contractor shall provide training via instructor-led training, interactive courseware, computer based training, and blended training solutions. During training events, the Contractor shall maintain attendance rosters and deliver training course completion certificates. Copies of the training rosters and training certificates shall be delivered to the PMO within 15 days of the completion of the training event.

Upon completion of training, the Contractor shall ensure students complete the Student Course Critiques (SCC) and Instructor Review Forms (IRFs) to evaluate training effectiveness and gather student feedback. A digital copy of all completed course attendee feedback, including SCCs and IRFs, shall be delivered to the Government PMO within 15 days of the completion of a training event. Within 30 days of completed training, the Contractor shall analyze training event critiques, identify issues and training effectiveness and provide recommendations and timeline for training improvement and training curriculum.

3.5.5 Training Device Sustainment

The Contractor shall create and provide a Gold Disk image with Release Notes containing all data for the current patch set/code release and identifying changes to the previous baseline. The Gold Disk image shall serve as the replica image to maintain all courseware environments consistent with the Production environment and product baseline, including MarineNet, GCSS-MC Portal, MTS, and DISA/GCDS.

The Contractor shall perform end-to-end support of the GCSS-MC/LCM MTS, including hardware and software maintenance, setup and loading of training material, software, and updates. The Contractor shall load the approved image and configure MTS current and future GEN servers and configure the MTS laptops. The Contractor shall maintain and update MTS/ customized menu scripts, processes, and procedures for data and environment backups/restore procedures.

The Contractor shall maintain the baseline and security posture of the R19 training environments.

The Contractor shall troubleshoot training device equipment requiring maintenance and conduct repair/replace actions including warranty actions as required.

Additionally, the Contractor shall troubleshoot the training device equipment requiring maintenance and conduct repair/replace actions including warranty actions as required.

3.6 PRODUCT LIFECYCLE SUPPORT

3.6.1 Reliability, Availability, Maintainability

GCSS-MC/LCM Increment 1 must deliver high quality services that achieve mission capability, satisfy user requirements and accomplish operation tasks in the most cost-effective manner. Reliability, Availability, and Maintainability (RAM) are key metrics used by the Government to evaluate the fielded system's performance, assess maintenance costs and support lifecycle sustainment planning. The Contractor shall measure and assess the system's performance against established RAM metrics and provide cost effective solutions and recommendations to achieve and maintain RAM objectives. The Contractor shall track and report RAM data at the weekly O&S meeting.

3.6.1.1 Reliability

System reliability is the average duration of hours that the system is operational for use, excluding scheduled downtime. GCSS-MC/LCM Increment 1 has an established reliability threshold is 99.44% with an objective of 99.99%. The Contractor shall collect data and use the Mean Time Between Failures (MTBF) and reliability formulas provided in

Figure 2 to compute the weekly MTBF and reliability percentage. The Contractor shall report events that result in deviations from the established threshold.

$$\text{Mean Time Between Failure (MTBF)} = \frac{\text{Total Time}}{\text{Number of Failures}}$$

$$\text{Reliability} = \frac{\text{Total Time} - (\text{Planned Downtime} + \text{Unplanned Downtime})}{\text{Total Time} - \text{Planned Downtime}}$$

Figure 2: Reliability Formula

Operational Availability (Ao) is a measure of degree to which an item is in an operable state and can be committed at the start of a mission when the mission is called for at an unknown point in time. The established threshold for operational availability is 95.00% with an objective operational availability of 98.00%. The Contractor shall collect the data and use the formula provided in

Figure 3 to compute the weekly Ao. The availability parameters shall include both corrective and preventive maintenance requirements. These measures do not include maintenance downtimes associated with issues beyond the scope of GCSS-MC/LCM control such as external networks and end-user devices.

$$\text{Operational Availability (A}_o\text{)} = \frac{\text{Total Up Time}}{\text{Total Time}}$$

Figure 3: Operational Availability Formula

3.6.1.2 Maintainability

Maintainability is the ability of an item to be retained in, or restored to, a specific condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair. The established threshold for GCSS-MC/LCM Increment 1 is no more than four (4) hours of system downtime for 90.00% of downtime occurrences. The Contractor shall compute and report the Mean Corrective Maintenance Time (MCMT) and Corrective Maintenance Downtime Occurrences (CMDO) using the formulas provided in Figure 4 below.

$$\text{Mean Corrective Maintenance Time (MCMT)} = \frac{\text{Total Unplanned Maintenance Time}}{\text{Total Unplanned Maintenance Actions}}$$

$$\text{Corrective Maintenance Downtime Occurrences} = \left(\frac{\text{Total Unplanned Maintenance Time}}{\text{Total Downtime}} \right) * 100$$

Figure 4: Maintainability Formula

3.6.2 Asset Tracking

The Contractor shall support Government asset management activities by providing accountability, tracking, and reporting of assigned hardware, software, licenses, and warranty information. The Contractor shall provide hardware, software, and firmware asset data elements as defined by the Government. The Contractor shall report any changes to asset data, such as changes in equipment location, resulting from warranty actions.

Upon the receipt of any Government Furnished Equipment/Information (GFE/GFI), the contractor shall inventory and complete a signed confirmation of the transfer for the property or information in the approved Government

system. The Contractor shall report all additions, changes, or deletions to an environment/location GFE/GFI, or if the responsible party changes. The Contractor shall notify the COR the next business day if they have received GFE/GFI that is defective, using the Contractor's own format while including information adequate to determine what GFE/GFI is defective and a description of the defect.

The Contractor shall support configuration status accounting, and physical configuration audits and physical inventories to maintain an accurate accountability of GCSS-MC/LCM Increment 1 assets.

The Contractor shall provide a semi-annual report on all Government Furnished Property.

Deliverable

CDRL A003 - Government Furnished Property Inventory Report

3.6.3 Licenses

The Contractor shall provide a quarterly license report with information to support licensing agreements including, but not limited to, number of licenses, users for named licenses, concurrent license users, servers, license unit, associated core processors, license renewal date, and license location. The Contractor shall provide cost effective licensing strategies that reduce total ownership costs for software licenses.

3.6.4 Warranty Support

The Contractor shall execute hardware and software warranty support agreements required to sustain and maintain the system. The Contractor shall track and report quarterly the status of all warranties and warranty actions. The Contractor shall assist the PMO in determining the most cost effective licensing strategy that reduces or minimizes lifecycle costs. This includes reporting trends in warranty actions to the PMO and using the data for lifecycle cost analysis that results in cost effective licensing strategy recommendations. The Contractor shall provide all support necessary and incidental to the tracking, procuring, and maintenance of system and system component warranties. This may require that the Contractor research and provide necessary technical and administrative information to the PMO or directly to hardware and software vendors. All warranty support efforts shall be performed in close coordination with program office representatives.

3.6.5 Obsolescence Management

The Contractor shall provide obsolescence management for all system components. The contractor shall assess hardware and software lifecycles and provide a comprehensive Hardware and Software Refresh Plan (CDRL G001) to mitigate obsolescence risks. The plan shall cover a rolling five (5) -year period for GCSS-MC/LCM Increment 1 assets. When directed by the PMO, the Contractor shall dispose of assets.

The Contractor shall notify the PMO of announced product end of life, loss or impending loss of manufacturers of items or suppliers of items, or software end-of-life. The Contractor shall provide systems recommendations for upgrades or migrations to mitigate obsolescence issues. The Contractor shall provide procurement and implementation costs to replace obsolete items and provide source data to support forecasting of obsolescence risks. The Contractor shall prepare hardware for disposal when required.

Deliverable

CDRL G001 - Open Systems Management Plan – (Hardware and Software Refresh Plan)

3.7 SERVICE TRANSITION

Service Transition encompasses the planning and execution of all system changes for GCSS-MC/LCM Increment 1. During this time, no new capabilities will be introduced. The only changes authorized will be for break fixes and to address IA/Cybersecurity requirements. Proposed system changes require a responsive, agile, yet disciplined approach to the GCSS-MC system and are categorized by components of the system and are aligned with the Program's organization:

- Infrastructure including hardware, operating system, and databases

- Functional business processes
- Cybersecurity System Security
- Data Quality/Interfaces

GCSS-MC/LCM Increment 1 currently has a documented backlog of approximately 624 CRs awaiting disposition. The CRs cover a broad area including functional, system security, interface management, infrastructure, training, audit, and WAN optimization. The CRs vary in level of priority (based on IEEE 12207 classification) and impact to the user community .

The GCSS-MC Configuration Control Board (CCB) reviews all CRs for the backlog. Each CR includes a change sub-type which identifies the GCSS-MC change authority (Change Assessment Board [CAB], CCB) and the associated configuration items impacted by the change. Once CRs are approved for the backlog, they are owned by their respective Product Owners. Over the course of FY2025, 1259 CRs were managed through the backlog. 361 CRs were applied to production during FY2025, averaging 90 CRs per quarter. Of the 361 CRs applied to production, 111 were for system cyber security and infrastructure, and 36 were for audit compliance. This schedule also accounts for the implementation of 5 (on average) quarterly Oracle CPU CRs to maintain system security compliance. During FY2025, 45 CRs were created from Incident and Problem Management from the Tier 1, 2, and 3 Service Desks. Although this was the previous activity for all CRs, we anticipate that the numbers will be lower during this sustainment effort since we will not be including any additional or future capability CRs in this process.

The Contractor shall provide responsive, disciplined support to sustain the GCSS-MC/LCM Increment 1 system following the Configuration Management Process detailed in Figure 6.

3.7.1 Change Requests

The Contractor shall participate in required change management boards (CABs, CCBs, and Release Management). The Contractor shall provide required Oracle status updates and other input as required for any change requests generated.

Upon request of the CAB, the Contractor shall provide a detailed technical approach, schedule, and cost for development and implementation of any identified change. Estimates shall include the time and resources required for the analysis, development, testing, deployment and documentation, and recording of approved changes in the authoritative data tracking tool. The Contractor shall provide support for problem management to include trouble ticket creation and management, trouble ticket investigation, RCA, detailed estimate analysis, and RICECPW impact assessments.

The Contractor shall identify requirements, design and implement the solution, and document the integration to meet technical data integration needs upon approval of the Government. The Contractor shall review their proposed designs and identify any integration inconsistencies prior to the submission of any change request.

The Contractor shall analyze integration requirements with external systems and design integration solutions and standards utilizing SOA technologies or workflow. The Contractor shall identify potential causes of issues occurring at the middleware layer (e.g., SOA/Fusion middleware) and provide corrective action to resolve.

The Contractor shall sustain existing GCSS-MC/LCM Increment 1 interfaces by inspecting, testing, analyzing, and repairing interfaces. The Contractor shall analyze system changes, conduct an impact assessment, and provide a level of effort to implement solutions in response to policy, cybersecurity, system, user requirements, or interface partner requirements. When necessary, the Contractor shall create and submit a CR to document the change and resulting analysis.

The Contractor shall, upon approval, create, test the technical solution, update documentation (CDRL D001), and implement approved system changes to correct identified system or application defects. The contractor shall provide unit test results to the Government as entrance criteria for test.

The Government will provide the Contractor with the prioritized list of CRs to be implemented for each Task Order period of performance. Upon agreement of the Task Order's scope of effort between the Government and the Contractor, the Contractor shall be responsible for implementing the CRs. If the proposed CR is estimated less than 100 man- hours to implement, it will be considered part of the basic sustainment of the PDSS system and there will be no additional funding allocated to the contractor for the execution of the approved change. All of the work involved in the identification, design, and presentation of a proposed solution are part of Core PDSS requirements under CLINS X001.

Deliverable

CDRL D001 - Technical Report - Study/Services - (Technical Data Package)

3.7.2 Engineering Change Proposals (ECPs)

ECPs are proposed changes or enhancements to the design, operation, and/or functionality of GCSS-MC/LCM Increment 1. ECPs may reference and include one or more CRs.

During this period, ECPs will only be considered for critical sustainment actions (e.g., addressing major obsolescence or security mandates) and are strictly prohibited for developing new features.

The Contractor shall provide responsive, disciplined support to maintain, change and enhance the system. The Contractor shall adhere to established Government processes and procedures in developing and implementing ECPs.

Upon request of the CAB, the Contractor shall provide a detailed technical approach, schedule, and cost to develop and implement the identified ECP. Estimates shall include the RCA, time and resources required for the analysis, development, testing, deployment and documentation, and recording of approved changes in the authoritative data tracking tool. The Government will review the contractor's ECP for acceptability and provide comments to the Contractor on the ECPs as necessary. All of the work involved in the development or implementation of an ECP are part of Core PDSS requirements under CLINS X001 if the total estimate is less than 100 man-hours combined and no additional funding will be allocated. Above 100 man-hours for the combined effort may result in a negotiated task order.

3.7.3 System Changes

The Contractor shall participate in required change management boards (CABs, CCBs, and Release Management). The Contractor shall provide required technical input for system changes, status updates of existing changes, and other input as required.

3.7.3.1 Change Analysis

The Contractor shall analyze the CR and provide the technical approach to implement the change. The Contractor shall provide a detailed estimate of time, required skills and cost for the change, to include analysis, building, test, documentation and recording in the authoritative tracking tool(s). For trouble tickets and issues identified by the Contractor, the Contractor shall create a CR using the Serena Business Manager CR Workflow for trouble ticket investigation, RCA (CDRL A001), detailed estimate analysis, and RICECPW impact assessments.

Deliverable

CDRL A001 – Scientific and Technical Report - (RCA Report)

3.7.3.2 Change Development

The Contractor shall develop configuration fixes, data scripts, and software for any correction of deficiencies, fixes, or changes to existing RICECPW within the sustainment of the current GCSS-MC/LCM Increment 1 baseline that

have been approved through the CM process. The Contractor shall generate RICECPW changes using PL/SQL, Java, Oracle Forms, Workflow, BPEL, XML Publisher, Discoverer/OBIEE, and Business Intelligence (BI) Publisher. The Contractor shall provide functional design for requests for change, identify and create tickets for technical deficiencies within the code base when discovered. The Contractor shall maintain the integrity of the technical solution throughout the change process. The Contractor shall provide all custom data, custom code, SQL and all original work product deliverables under this task order in a non-proprietary format.

The Contractor shall check in all system changes into the code repository and provide code migration documentation and/or procedures to the Government for Independent Verification and Validation (IV&V).

The Contractor shall ensure the code being developed and released to the Government meets the requirements for a secure baseline. The Contractor shall support all IA/Cybersecurity activities required to operate, maintain, and sustain the SDE. The Contractor shall manage all aspects and associated relationships of the SDE, and provide both physical and remote access to the Government.

3.7.3.3 Change Functional Testing

The Contractor shall conduct unit testing and functional testing in the development environment to verify that patches and fixes successfully resolve identified issues. The Contractor shall provide GCSS-MC/LCM Increment 1 functional application support including functional application testing. The Contractor shall perform backup and recovery testing and execute regression testing, as required. The Contractor shall create or modify existing test cases and scripts to test changes. The Contractor shall create and maintain regression tests for automated regression testing. The Contractor shall participate in the retest of current requests for change.

The Contractor shall provide unit testing results for Independent Verification and Validation (IV&V).

3.7.3.4 Change Documentation Activities

The Contractor shall create and/or update technical baseline documents (CDRL D001) directly impacted by the change and submit for review. Upon completion of the review, the Contractor shall incorporate comments and maintain the documentation via the documentation and CM approval processes.

Deliverables

CDRL D001 - Technical Report - Study/Services - (Technical Data Package)

3.7.3.5 Change Promotion

The Contractor shall promote approved changes to the PSS and PROD following established GCSS-MC/LCM Increment 1 release processes.

3.8 SYSTEMS ENGINEERING SUPPORTING ACTIVITIES

3.8.1 Systems Engineering

GCSS-MC/LCM Increment 1 is a fielded system in the O&S phase as defined in DoD Instruction 5000.02. The primary O&S systems engineering objectives are to maintain the system by identifying and resolving root causes of issues and sustaining the system through upgrades and technical refresh - all in a cost effective manner.

The GCSS-MC Program Office's systems engineering activities align with DoD, DON and Marine Corps engineering regulations and processes as defined in DoD Directive 5000.01, "The Defense Acquisition System" and Marine Corps Systems Command Order 5400.5, "Systems Engineering Technical Review (SETR) process." Additionally, the PMO has developed a Systems Engineering Plan (SEP) to guide O&S efforts. The purpose of the SEP is fourfold:

- 1) Establish the operational activities that must be performed to sustain the current and future-state system;
- 2) Define the future-state of the GCSS-MC/LCM Incr 1 PDSS system;

- 3) Rationalize and arrange the identified technical initiatives that will be executed to achieve the future-state system;
- 4) Describe the engineering processes the PMO will execute to sustain that future-state system.

Systems engineering provides an approach and framework for product sustainment and development in a system context. It encompasses what all organizational entities and all enterprise personnel must accomplish in order to produce a quality, competitive product that will provide an acceptable return on investment to the enterprise and will achieve customer satisfaction. Systems engineering integrates the multiple engineering disciplines of systems engineering, software engineering, configuration management, information assurance/cybersecurity, architecture, interface and data qualities management and testing to deliver a well-designed system. The SE process defines the requirements for the total technical effort related to development of products and processes that will provide lifecycle support. Systems engineering is applied recursively to the development or incremental improvement and simultaneously provides related lifecycle processes for product development, test, operation, support, training and disposal. PDSS for GCSS-MC/LCM Increment 1 includes applying the systems engineering “V” model, cybersecurity techniques which maintain system security and compliance with IA policy, and technical management processes identified below, integrated with ITIL v3 Service Transition and Service Operations processes.

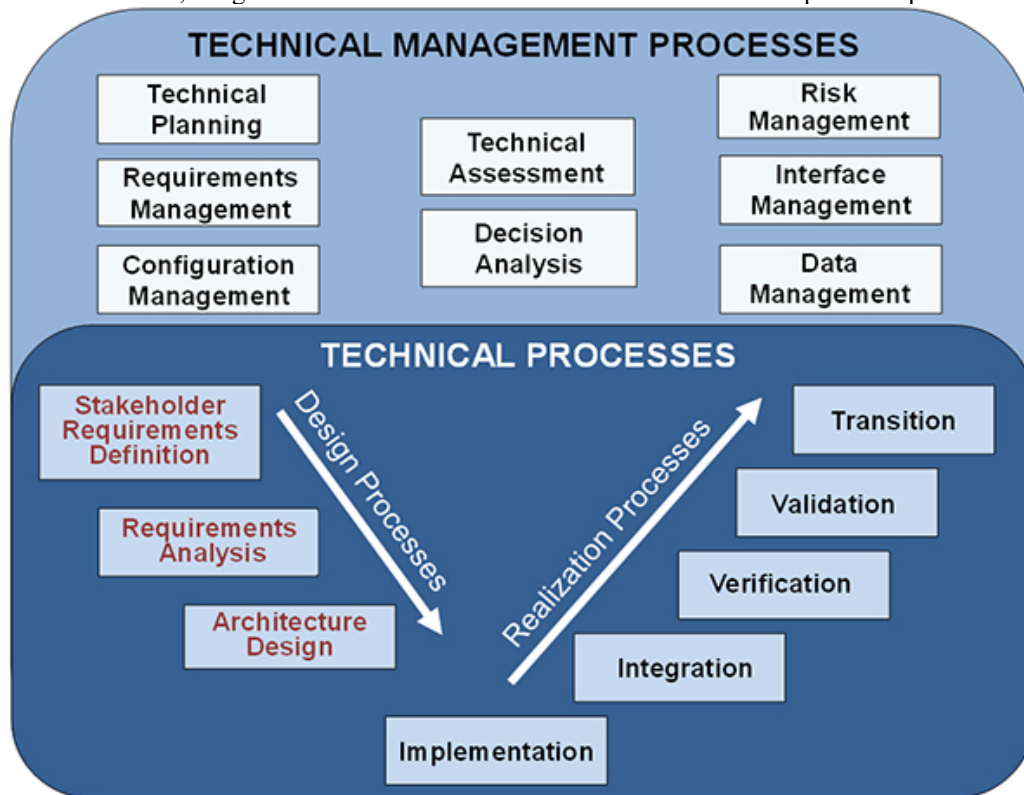


Figure 5: Systems Engineering Process

The Contractor shall provide disciplined systems engineering and apply acceptable principles, criteria, and procedures to all GCSS-MC/LCM Increment 1 tasks in compliance with DoD, DON, Marine Corps, and program processes.

3.8.1.1 Sustainment Engineering Processes

The PMO has identified 12 processes critical to sustainment engineering. These processes define the process flow, decision points, and actions to be followed in executing sustainment engineering. The Contractor shall integrate their processes into the PMO processes and provide recommendations to enhance or streamline processes.

3.8.1.2 Systems Engineering Management Plan

The Contractor shall provide a Systems Engineering Management Plan (SEMP) (CDRL H001) that describes the Contractor's overall systems engineering management approach. The SEMF shall describe the standards, capability models, processes, and toolsets the Contractor will use in executing PDSS. The Contractor shall describe how they will align their SE processes to the Government's processes. Specific areas to be addressed in the SEMF include:

- 1) Organization of the team, including physical location and facilities to be used
- 2) Management approach and use of the GCSS-MC/LCM Increment 1 environments
- 3) Description on use of engineering standards, capability models and tools to be employed
- 4) Technical risk management
- 5) Requirements management and traceability
- 6) Configuration management process for identifying and managing configuration items
- 7) Information Assurance/Cybersecurity processes and tools
- 8) Test process and methodology
- 9) Software change methodology
- 10) Build management process
- 11) Architecture/System Design process
- 12) Data Management processes
- 13) Documentation standards to be used and management approach
- 14) Production support strategy
- 15) Change support strategy

The Contractor shall describe their approach to managing resources if personnel are shared between production and change teams.

Deliverable

CDRL H001 - System Engineering Management Plan (SEMF)

3.8.1.3 Systems Engineering Support

The Contractor shall provide traceability of CRs through design, development, test, and deployment of system updates.

The Contractor shall provide recommendations, including analysis and technical risk assessment for all system incidents, problems, updates, and changes.

The Contractor shall maintain the existing solution architecture (logical, physical, and network topology) and system diagrams. As changes occur, the Contractor shall provide updates to system diagrams and architectural drawings to reflect changes to the baseline architecture to maintain the integrity of the technical solution.

The Contractor shall identify and manage technical risks, work with the Government to capture and quantify the potential impact of the risk, assist in defining mitigation approaches, and participate in risk reviews.

The Contractor shall support data calls and data extracts required for cybersecurity certifications.

3.8.1.4 Systems Engineering Tool Support

The Contractor shall implement SE tools to support engineering processes; however, all tools must be compatible with those used by the GCSS-MC PMO.

The PMO currently uses a variety of tools to manage GCSS-MC/LCM Increment 1 efforts. This includes: ~~IBM® Rational® DOORS® for Requirements Management~~; Serena® Business Manager for Configuration and Asset Management; Hewlett Packard Application Lifecycle Management (HP ALM®) for Testing; Oracle Enterprise Manager/Business Application Manager and Serena Dimensions CM® for software development. The Government has established a SLA to host these tools in the Naval Systems Engineering Resource Center (NSERC). The

Contractor may implement systems engineering tools of their own choosing to support internal engineering processes. However, they must be compatible with those used by the GCSS-MC PMO.

The Contractor shall provide user account and database administration of the CM, asset management, and test tools as requested by the Government. The Contractor shall provide user notifications of outages and work with NSERC personnel to identify performance issues. The Contractor shall provide recommendations to enhance the use of the capabilities of these tools, including integration with other tools. If the tool's hosting provider changes, the Contractor shall provide recommendations for the transition of the tool and associated data to the new provider. The Contractor shall maintain a record of requested updates to the tools and administer SE tools working group meetings to prioritize and decompose requirements for requested updates. The Contractor shall maintain access lists and configure users for system access, once access is granted. The Contractor shall implement tools and methods that provide a complete record of test documentation.

The Contractor shall provide requirements management data that is compatible with the schema resident in the interim GCSS-MC/LCM Increment 1 SharePoint site and shall transfer the data to a permanent repository as directed by the Government. The GCSS-MC PMO also uses MS SharePoint for management of technical data. A GCSS-MC Data Repository exists in the Marine Corps' SharePoint Online. The Contractor shall provide all CDRLs electronically to the CM data repository including documentation to support system changes. The Contractor shall review and update data in the GCSS-MC data repository to support PDSS IPTs and Working Groups.

The PMO uses BMC ITSM for the Enterprise Service Desk. The Contractor shall support the Government's administration of Remedy by providing recommended improvements, integration opportunities with other SE tools used by GCSS-MC, as well as training tools. The Contractor shall use a tool that is compatible with Fortify for the automated exchange of code scan findings with the GCSS-MC PMO. The Contractor shall provide all data and information in a format compatible with PMO established tools and processes.

3.8.2 Software Engineering

The GCSS-MC PMO requires disciplined software engineering as an integral component of the overarching systems engineering approach needed to sustain the GCSS-MC Increment 1 system. The Contractor shall employ the use of sound software engineering principles, methods, practices, and tools to produce and sustain reliable GCSS-MC/LCM Increment 1 applications .

GCSS-MC/LCM Increment 1 currently employs a modified agile development methodology for application sustainment. However, the Contractor may employ the approach that best meets the requirements to deliver reliable, high quality and cost effective software solutions. The Contractor shall perform disciplined application development activities consistent with CMMI Level 3.

3.8.2.1 Software Development Plan

The Contractor shall provide a Software Development Plan (SDP) (CDRL H002). The SDP shall describe how the Contractor will align their software engineering processes to the Government's processes. Specific areas to address in the SDP include:

- 1) Plan introduction and overview
 - Purpose, scope, and objectives
 - Assumptions and constraints
- 2) Relationship to other program plans
- 3) Referenced documents
- 4) Identification of all software and software products to which the SDP applies
- 5) Definition of terms and acronyms
- 6) System overview, including system and software architecture
- 7) Overview of required work, including:
 - Requirements and constraints on the system and software to be developed
 - Software products and related deliverables
 - Requirements and constraints on project documentation

- Additional requirements and constraints such as on project security, privacy, methods, standards, interdependencies in hardware and software development.
 - Known software-specific risks
- 8) Project organization and resources
 - 9) Plans for performing software change activities, including:
 - Software change processes
 - Software change approaches/development approaches
 - Software change methods
 - Software change standards
 - Reusable software products and COTS
 - Software types/categories (i.e., operational software, test software, support equipment software) and associated processes, controls, and documentation.
 - Handling of critical attributes (such as safety, security, and information assurance)
 - Computer resources utilization and reserve capacity/growth management
 - 10) Supporting processes and information, including:
 - Software risk management
 - Software quality assurance
 - Approach to requirements traceability

The Contractor shall implement application solutions which sustain code and system configurations and associated documentation to support correction of defects and enhancements.

Deliverable

CDRL H002 - Software Development Plan (SDP)

3.8.3 Interface and Data Management

3.8.3.1 Interface Management

Interface Management is responsible for identifying, implementing, and maintaining the internal and external interfaces necessary for system operation. Interface Management ensures a common interface definition and efficient use and compliance among the system elements as well as with other systems.

GCSS-MC/LCM Increment 1 exchanges information via more than 30 interfaces across 20+ external parties in all GCSS-MC environments. The current list of interfaces can be found in SDE Technical Data Package. This frequently involves extensive global coordination. The Contractor shall execute full lifecycle management (implement, support, maintain, modify, enhance, and decommission) for GCSS-MC/LCM Increment 1 interfaces.

The Contractor shall define, control, and communicate the information needed to enable unrelated objects (i.e., systems, services, equipment, software, and data) to interoperate cohesively. The Contractor shall support the control measures (Configuration Management) to ensure that all internal and external interface requirement changes are properly documented.

The Contractor shall define, design, and manage the physical and logical interfaces, both internal (communications between system elements) and external (communications between the system and other systems), to ensure the interfaces perform correctly.

The Contractor shall support the definition, implementation, and testing of internal and external interfaces, including any associated dependencies, to ensure that systems operate as designed and meet stakeholder requirements throughout the life cycle.

The Contractor shall work closely and cooperatively with internal and external organizations and agencies to manage agreements between GCSS-MC/LCM Increment 1 and other systems owners. The Contractor shall follow established processes and procedures, as documented in Interface Control Documents (ICDs) that capture the

technical agreement between the GCSS-MC PMO and the partner system owner, including interface design, data integrity, network, and infrastructure.

The Contractor shall organize and lead the monthly GCSS-MC Systems Integration IPT (SI IPT) and provide status and system integration information, as requested, to support PMO briefs. At the SI IPT, the Contractor shall identify, communicate, discuss, plan, and execute approaches toward avoiding or resolving issues related to interfaces between GCSS-MC/LCM and external systems. The Contractor shall deliver Meeting Minutes and Action Items capturing IPT highlights and action items.

3.8.3.2 Data Management

The Contractor shall evaluate changes in policies and compliancy to assess impacts to the Increment 1 and external partners, provide recommendations and develop solutions to address change impacts. The Contractor shall maintain and update documentation, including the Data Management SOP and TA.155:PROD24 (Data Maintenance).

The Contractor shall analyze Defense Logistics Management Standards (DLMS) changes and define technical impacts to the system and the data exchanges.

The Contractor shall provide Data Management Planning to identify impacts to the system environments.

3.8.4 Information Security Management/Cybersecurity

The objective of cybersecurity is to protect Marine Corps information from internal and external threats and attacks, while ensuring the confidentiality, integrity, and availability of information.

To achieve cybersecurity objectives, the Contractor shall adhere to the requirements of Marine Corps Order (MCO) 5239.2B and DoD Instruction 8510.01, "Risk Management Framework (RMF) for DoD IT" for GCSS-MC/LCM Increment 1, and ensure that applicable personnel are certified in accordance with Appendix 4 in SECNAV M-5239.2 Cyberspace Information Technology and Cybersecurity Workforce Management and Qualification Manual. The Contractor shall support all assessment and authorization activities throughout the system life cycle in accordance with the latest releases or revisions of the cybersecurity policies.

The Contractor shall participate in cybersecurity and vulnerability assessment scan reviews and provide technical guidance and solutions implementing cybersecurity best practices which will increase the security of the system and mitigate or eliminate vulnerabilities. The technical guidance and solutions must align with applicable STIGs.

The Contractor shall apply security updates and patches to software and operating systems and conduct verification testing. The Contractor shall document all findings in a weekly cybersecurity status report.

The Contractor shall review IAVAs, Information Assurance Vulnerability Bulletins (IAVBs), Technical Advisories, Communications Tasking Orders, Marine Corps Directives, Operational Directives, CPUs, vulnerability alerts, and vendor notifications to determine applicability to GCSS-MC/LCM Increment 1 and to assess impact and provide assessment to the ISSM. The Contractor will track, report status and provide remediation recommendations for the vulnerabilities.

The Contractor shall provide an impact assessment of CPUs if there is an impact to RICECPW and document the assessment in CR. This impact analysis shall include a detailed estimate to modify the RICECPW element for compatibility with the impacted CPUs. The impact assessment shall indicate if no changes are required.

The Contractor shall maintain a Security Plan of Actions & Milestones (POAM) (CDRL E001) that lists all vulnerabilities identified by every assessment that identifies vulnerability.

The Contractor shall provide encrypted RICECPW code to the GCSS-MC ISSM for code review when updates occur. When the Contractor receives the code review results from the GCSS-MC ISSM, the Contractor shall fix issues found or provide proposed justification for not fixing.

The Contractor shall support all activities required for maintaining the ATO or Interim Authority To Test (IATT) and remaining compliant with the Federal Information Security Modernization Act (FISMA). These activities include, but are not limited to: Annual Security Reviews, Annual Security Control testing, Annual Contingency Plan testing, and update and submission of a quarterly Security POAM (CDRL E001).

The Contractor shall support Cyber Readiness Inspection and IV&V events as required by the GCSS-MC ISSM. This task includes, but is not limited to: reviewing and updating systems security documentation, performing pre-assessment scans, analyzing vulnerability scan results, analyzing and updating configuration documentation, evaluating STIGs, evaluating test results, preparing and reviewing a Security POAM (CDRL E001), providing remediation options for vulnerabilities, and remediating the system. All vulnerabilities shall be identified in the Security POAM.

The Contractor shall update GCSS-MC/LCM Increment 1 cybersecurity documentation in accordance with DoD policy and upload that documentation to a location identified by the ISSM where it is accessible to authorized individuals.

The Contractor shall generate, review, and update cybersecurity documentation as required by DoD RMF processes.

The Contractor shall support cybersecurity testing by maintaining a Cybersecurity Detailed Test Plan (DTP) (CDRL E002) that identifies the specific system tests and test procedures to be performed and all STIGs used.

The Contractor shall support cybersecurity testing by providing a thorough Cybersecurity Risk Assessment that identifies the security posture of the system after a test. The Contractor shall identify any procedures not performed and detail the reason for non-compliance with pre-established DTP.

Deliverables

CDRL E001 – Technical Report - Study/Services – (Security Plan of Actions and Milestones (POAM))

CDRL E002 – Test Plan – (Cybersecurity Detailed Test Plan (DTP))

3.8.5 Configuration Management

A key component of GCSS-MC/LCM Increment 1 systems engineering is the planning, assessing, analyzing, and controlling of various technical activities. Configuration Management (CM) is a technical process within systems engineering which enables technical control of program activities, system design, work products, requirements, interfaces, data and configurations; and changes to these, affecting cost, performance and schedule. For large enterprise information systems, the DoD has adopted the ITSM model. ITSM employs ITIL v3 documented best practices. ITIL includes CM as part of service transition, also referred to as transition planning and support.

The Contractor shall provide, implement, and adhere to a CMP (CDRL D002) that outlines the Contractor's strategy for identifying, creating, recording, reporting, changing, controlling, and managing all GCSS-MC/LCM Increment 1 Configuration Items (CIs). Configuration Items shall encompass all GCSS-MC/LCM Increment 1 hardware, software, firmware, documentation, and their versions, statuses, relationships, configurations, and baselines.

The Contractor shall follow the PMO's configuration control and approval processes required to change a CI's attributes and re-baseline the CIs. The Contractor shall ensure all CRs properly document affected CIs.

The Contractor shall ensure that releases are coordinated with change management and release management. The Contractor shall provide a Product Baseline (CM.070) for each patch set. The TDP includes all updated GCSS-MC/LCM Increment 1 CIs.

The Contractor shall establish and implement configuration control and approval processes required to change a CI's attributes, to re-baseline the CIs, and then report the CI completion.

Major releases to the production system are performed on a periodic basis. Each major release is comprised of a TDP which contains a complete copy of the deployed production software, the database, and all configurations. The TDP is the CI for that release and is stored in the CM Data Repository on SharePoint. The TDP of a major release must be able to install an entirely new system from the ground up, without having to run incremental installation updates.

Point releases, performed at regular intervals between major releases, consist of software updates, code files, database scripts, configuration scripts, and installation instructions as necessary to update the current production system. The point release is encapsulated within a TDP. After several point releases, a major release aggregates all point releases, since the last major release, and deploys one package.

The Contractor shall follow established change, documentation and release management processes, and describe their approach to accomplishing the activities. The Contractor shall integrate their CM processes into the Government's CM processes in accordance with the GCSS-MC/LCM Increment 1 CMP and government SOPs. System changes are driven by many factors, including DoD, DON, and U.S. Marine Corps (USMC) policy. Policies include but are not limited to logistics and cybersecurity, fiscal compliance requirements, user requirements, and technology changes. System updates are implemented using a combination of CRs and ECPs managed through the GCSS-MC PMO configuration management process. CRs are used for routine changes required to sustain the system. System changes will be managed in accordance with the GCSS-MC/LCM Increment 1 SEP, CMP, SETR process, and applicable PMO plans and processes, which can be found in the SDE Technical Data Package.

A backlog of system changes exists. Today the PDSS provider for GCSS-MC/LCM Increment 1 releases updates to the Production system as fixes become available. The schedule also accounts for quarterly CPUs from Oracle. The Contractor will support a quarterly release cycle.

Deliverable

CDRL D002 – Supplier's Configuration Management Plan

3.8.5.1 Configuration Management Process

Figure 6 depicts a high level view of the GCSS-MC/LCM Increment 1 CM process. The Contractor shall support and integrate CM processes into the Government's processes.

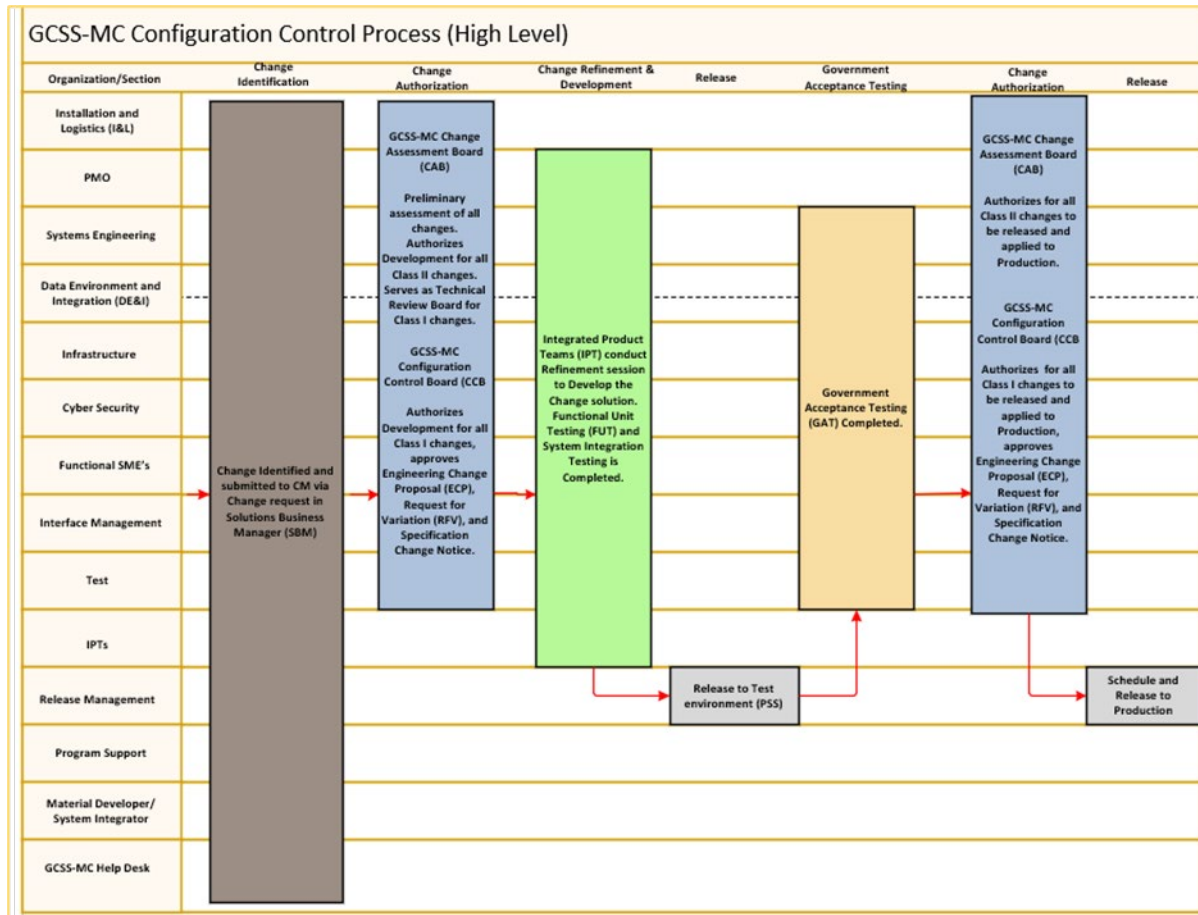


Figure 6: GCSS-MC Configuration Management Process

3.8.5.2 Configuration Control Board (CCB)

The Contractor shall support the CR change sponsor at CCBs to discuss the proposed change and technical solution. The Contractor shall prepare CCB briefs as required to address proposed solutions and courses of action (COAs). The Contractor shall ensure necessary technical representation is present to address proposed solutions and COAs.

3.8.5.3 Change Management

The Contractor shall adhere to the PMO's configuration control processes required to change a CI's attributes and re-baseline the CIs. The Contractor shall follow the PMO's CR workflow and ensure approved changes are fully documented throughout the CR life cycle.

3.8.5.4 Change Assessment Board (CAB)

The Contractor shall participate in CABs and present their evaluation of approved CRs, including a technical assessment of priority and severity, recommended technical approach based on root cause to correct or implement a design change, affected documentation, scope of effort, and total cost to implement the change, including impacts to training. The Contractor shall identify opportunities to implement block upgrades for related CRs where implementing block upgrades can save time, money, or enhance system performance. The Contractor shall deliver required artifacts to the GCSS-MC Data Repository and Code Management repository.

3.8.5.5 Configuration Identification

Upon approval by the CAB and CCB, the Contactor shall identify, decommission, and revise configuration items in the Configuration Management Database (CMDB). The Contractor shall produce Configuration Status Accounting (CSA) reports to identify software and hardware baselines (CDRL D003).

The Contractor shall support PMO Physical Configuration Audits (PCA) by locking CIs in the CMDB. The Government conducts PCAs at least annually. The Contractor shall review and provide inputs to PCA plans and draft PCA reports.

Deliverable

CDRL D003 - Configuration Status Accounting (CSA) Information

3.8.5.6 Code Management

The Contractor shall describe their approach for code management in the CMP, to include the processes for code migration, deployment of software code, and configuration control, including object versioning, for developing, managing, and deploying GCSS-MC/LCM Increment 1 software code.

The Contractor shall maintain the integrity of the software development and CM controlled environments, ensure consistent implementation of CAB and CCB decisions, and enable efficient processing of hardware changes, code fixes, data fixes, configuration fixes, design changes, test scripts, and test configurations.

The Contractor shall coordinate migration and delivery of Software Builds to the GOC for promotion to the PSS and PROD environments upon authorization. As new builds and releases are developed, the Contractor shall provide installation and release notes, in conjunction with the code builds, and post the builds to the Government's designated data repository. The Contractor shall migrate the quarterly build code to the designated Government Code Repository using the established file structure. Based on Government direction, the Contractor shall freeze the production baseline and create a patch set with the associated documents for the Product Baseline (CM.070).

3.8.5.7 Release Management

The system's release schedule involves planning, approval for system outages from stakeholders, and management of the resources required to execute the schedule. A quarterly release schedule is required to meet operational availability requirements.

The Contractor shall attend the monthly PMO release planning meeting and provide a proposed release plan identifying CRs for appropriate releases based on the problem management backlog assignments and PMO ranking. The Contractor shall provide status updates for ongoing release plans at the release planning meeting.

The Contractor shall schedule, coordinate, plan, and implement approved releases with CM. The Contractor shall assign builds to appropriate software releases as identified in the GCSS-MC CMP and develop 30/60/90 day release schedules based upon assignment, prioritization, and scheduling, including refreshing the environments defined in Section 1.1.

The Contractor shall maintain a PSS and Production Software Baseline Reports (PSBR) and track CRs or software releases by build per release, patch set, and clones provided as GFI.

The Contractor shall gather approved CR and build-specific data by attending CAB and CCB meetings, Test Incident Reporting (TIR) meetings, testing hot-washes and applicable meetings to incorporate into an assignment, prioritization, and scheduling spreadsheet.

The Contractor shall coordinate with the Government to schedule ad hoc system backups and database clones for promotion to solution development and test instances, and compile the 30/60/90 day release schedule. The Contractor shall prepare DD Form 1149 and track distribution of clones and clone contents (builds).

The Contractor shall identify requirements and propose an associated schedule for technical refresh of these environments.

3.8.6 Validation and Testing

The Contractor shall perform service validation and testing in conjunction with the development and deployment of system changes and in accordance with the GCSS-MC engineering processes. Validation and testing includes test environment and tools, test planning, external partner requirements, test execution, and test reporting.

The Contractor shall provide a test schedule within the Detailed Test Plan identifying test events and activities as required by the Government. When required by the Government, the Contractor shall participate in test meetings hosted by the Government, such as test planning, test team, CCB, and deficiency review board meetings.

The Government retains the right to participate in or witness all contractor testing activities. The Contractor shall ensure Government access to all development, test, and PSS environments.

3.8.6.1 Test Environment

The Contractor shall perform test environment setup and coordinate unit testing, integration testing, system testing, regression testing, and performance/load testing. As required, the Contractor shall create a CR for test configurations and data scripts for test events.

The Contractor shall use the software development environment to perform unit testing. Upon successful completion of unit testing, the Contractor shall provide test results for IV&V. The Contractor shall maintain CM of the code used for System Integration Testing (SIT) and GAT and coordinate Government involvement to perform SIT. Following successful completion of SIT, the Contractor shall assist in configuring the PSS for GAT. The Contractor shall identify requirements and associated schedule for technical refresh of these environments as part of the maintenance function.

The Contractor shall maximize use of existing GCSS-MC facilities, software tools, and other Government developmental and integrated resources to reduce costs. The Contractor shall produce a report using GFI provided HP ALM when directed by the Government. The Contractor shall implement tools and methods that provide a complete detailed record of test documentation and provide data compatible with Government systems and accessible to Government designated personnel.

The Contractor shall implement tools and methods that provide a complete record of test documentation.

The Contractor shall identify and ensure test environments contain required test data necessary to facilitate operationally relevant (realistic) test and evaluation. The Contractor shall include procedures for test data cleansing, conversion, and verification/validation, as necessary. This data shall be transferrable and in a form usable by the Government.

3.8.6.2 Test Planning

The Contractor shall ensure all test documentation and activities are traceable to requirements, issues or incidents, including test plans, test cases, test data staging, test scenarios, and test scripts in accordance with established PDSS processes. The Contractor shall properly set up the test environment and required data to support testing. All test documentation is subject to Government approval prior to use in support of system changes, no matter the Contractor's perceived magnitude of the change. The Contractor shall perform RCA (CDRL A001) and may be required to provide a short and long term testing plan.

The Contractor shall plan and execute required test and evaluation processes to include data quality and performance tests as required. The Contractor shall provide test strategy, planning and execution schedule with external partners. The Contractor shall follow a repeatable test strategy for system changes. The Contractor shall identify and communicate test risks, resource requirements, supporting service needs, and scheduling milestones.

When required, the Contractor shall prepare and deliver for Government approval, test documentation, including detailed Test Plans (CDRL C001), test cases (CDRL C002), and test scripts (CDRL C003) that are traceable to requirements, issues or incidents. As a part of CDRL C001, the Contractor shall identify required Government support such as Government furnished property, equipment, information, external partners, or data required to conduct testing activities necessary to carry out any tasks in this section. The Contractor shall provide proposed test scripts and technical justification for level of testing, to include: function unit test, regression, and failover. The Contractor is encouraged to utilize prescanned scripts to resolve known issues. The Contractor shall provide a roll-back plan for each system change. All test documentation shall be uploaded by the Contractor into the Government's data repository.

Deliverables

CDRL A001 – Scientific and Technical Report - (RCA Report)

CDRL C001 - Test Plan

CDRL C002 - Software Test Description (STD) - (Test Case)

CDRL C003 - Software Test Description (STD) - (Test Script)

3.8.6.3 Test Execution

The Contractor shall execute tests in accordance with the Government-approved Detailed Test Plan.

For designated Contractor test events, the Contractor shall submit exploratory or incremental test results reflecting all completed test scripts, remaining test scripts, number passed, number failed, and status of failed tests.

The Contractor shall conduct SIT with the Government to demonstrate successful development testing results prior to promotion to PSS for GAT.

Following successful completion of SIT, the Contractor shall set up PSS and coordinate GAT. The Contractor shall monitor GAT and resolve environment, configuration, and code issues.

When required, the Contractor shall develop and collect TIRs or respond to Government provided TIRs, perform RCA, and provide analysis and solutions to resolve TIRs. The Contractor shall resolve established TIRs as defined in the DTP or applicable process.

The Contractor shall implement the roll-back strategy for all failed system changes.

The Contractor shall provide a final test report (CDRL C004) and update the CM process tool upon completion of the final system change test event. The Contractor shall validate test scenarios and test scripts which provide verification and validation of the GCSS-MC system functionality and operational performance parameters; document observation points of system anomalies and data collection plans; document technical requirements verification; analyze proposed work arounds and solutions to the root cause; and propose solutions to reduce recurrence of repetitive problems.

Deliverables

CDRL C004 – Software Test Report (STR)

3.9 FUTURE INITIATIVES

The elements in section 3.9 are meant to advise the contractor of the types of initiatives they may have to support in the future. If the initiative is already complete by the time of contract award, the contractor will be required to support the implemented initiative. If the initiative is not implemented and required by the Government, a separate task order may be negotiated and awarded to the contractor. The contractor shall have the ability to perform any of the tasks under this section, when ordered via task order.

3.9.4 Audit Compliance

The Government is responsible for improving program financial efficiency and accountability of system internal controls, business processes and supporting documentation of financial statements. The Contractor shall describe their approach and use of tools and processes to implement and sustain audit compliance. The Contractor shall enable GCSS-MC/LCM Increment 1 to perform audit readiness and reporting, including the ability to trace budgetary transactions to financial statements (capture and retain transaction data). Describe approach to support Implementation of (2) CPD Requirements Asset Valuation and Asset Depreciation using Oracle modules. Describe the approach for supporting Governance, Risk, Compliance Tool (GRC) and implementation to the GCSS-MC/LCM Increment 1 system. Support for Audit compliance and financial audit of GCSS-MC System support to include remediation activities associated with findings, requests and observations.

3.9.1

3.10 PROGRAM MANAGEMENT

The Contractor shall provide program management that supports planning, administration, execution, reporting, and continuous improvement of cost, schedule, and performance to satisfy the requirements identified in the PWS. The Contractor shall ensure tools used to manage, deliver, or support GCSS-MC/LCM Increment 1 are interoperable with PMO tools.

3.10.1 Post Award Conference and Project Kick-Off Meeting

The Contractor shall schedule a combined post award conference (PAC) and Project Kick-Off (PKO) meeting within thirty (30) calendar days of contract award at a Government designated facility in the Quantico area. The meeting provides the opportunity to discuss technical, management, security administration, travel authorization process, and reporting procedures. As part of the PAC/PKO, the Contractor shall introduce key personnel and present program implementation processes to be used under the contract, and this includes the current personnel staffing process and status and project organizational structure. Contractor shall provide an overview of their risk management process. The Contractor shall deliver a project kickoff brief at least 3 business days before the PAC/PKO. The brief will include: key risks, dependencies and mitigation for each identified risk. At a minimum, the attendees will include key contractor personnel, representatives from the PMO, other relevant Government personnel, and the COR. At a minimum, the Contractor shall provide the following at the PKO meeting:

- Transition-In Plan (Draft) (CDRL B003)
- Program Management Plan (Draft) (CDRL B001)
- Quality Management Plan (Draft) (CDRL B002)

Deliverables

CDRL B003 - Transition Plan - (Transition-In Plan)

CDRL B001 - Program Management Plan

CDRL B002 - Quality Assurance Program Plan - (Quality Management Plan)

3.10.2 Program Management Plan (PMP)

Within 30 calendar days after completion of the PAC, the Contractor shall establish a comprehensive PMP) describing the processes and procedures associated with how the project will be managed and executed, and ensures accurate and timely cost, schedule and performance information throughout the life cycle of the project.

The Contractor shall deliver a PMP (CDRL B001). The PMP shall describe the Contractor's approach towards:

- Managing and executing the Task Orders
- Identifying, monitoring, controlling, mitigating, and reporting risks
- Managing communications between the Contractor and the Government, its affiliates, and stakeholders
- Developing and maintaining SOPs for all tasks
- Developing and maintain an overall work breakdown structure (WBS)
- Developing and maintaining project cost, schedule and performance metrics
- Developing assumptions and constraints

Deliverable

B001 - Program Management Plan

3.10.3 Risk Management

The Contractor shall implement risk management processes to identify and mitigate project and technical risks and provide metrics to monitor project status. The Contractor shall participate with the Government to review and collaborate on risk assessments and mitigation approaches in accordance with the DoD Risk, Issue, and Opportunity Management Guide for Defense Acquisition Programs, dated June 2015, and the GCSS-MC Risk, Issue, and Opportunity Management Plan.

3.10.4 PDSS Schedule Management

The Contractor shall provide their schedule management approach to track and report activities required to operate and sustain GCSS-MC/LCM Increment 1, including program transition, configuration management, system changes, system outages, release management, system outages, cybersecurity activities, capacity management, asset management, backup management, and training. The approach shall also detail the Contractor's methods to develop, management, and report IMSs, as required through Task Orders.

3.10.5 Release Management Schedule

In conjunction with requirements of Section 3.8.5.7, Release Management, the Contractor shall deliver, as required, a Release Management Schedule (RMS) that provides a graphical representation of planned and actual start and finish dates for in-progress and upcoming GCSS-MC/LCM Increment 1 system releases, test activities, system outages, cybersecurity activities, and other events that may affect system availability. The Contractor's delivery of the RMS shall include a view of release management activities planned to occur within 90 days of delivery, as well as a multi-year view of release activities, inclusive of executed contract options and Task Orders throughout the contract's lifespan.

3.10.6 Contract Cost Reporting

Actual cost data will for CLIN 0001 will be delivered on a monthly basis and broken down to by major PWS elements, 3.1, 3.2, 3.3, 3.5, 3.6, 3.7, 3.8 and 3.10. This data will be compared to CLIN 0001, 1001, 2001, 3001 and 4001 for determination of the profit sharing amount.

3.10.7 Quality Management Plan

The Contractor shall provide a Quality Management Plan (QMP) (CDRL B002) that addresses the Contractor's approach to meet quality and performance standards outlined in the Performance Requirements Summary (PRS) of the Quality Assurance Surveillance Plan (QASP). The QMP shall include, at a minimum, the following components:

- Quality objectives
- Key project deliverables and processes to be reviewed for satisfactory quality level
- Quality standards
- Quality control and assurance activities, including process for reporting quality control and assurance problems
- Quality roles and responsibilities
- Quality tools

Deliverable

CDRL B002 - Quality Assurance Program Plan - (Quality Management Plan)

3.10.8 Performance Evaluation/Technical Status Meeting

When directed by the Government, the Contractor's Program Manager (CPM) shall convene periodic contract activity and status meetings with the COR and other designated Government stakeholders. The purpose of this meeting is to ensure all stakeholders are informed of activities, provide opportunities to identify other activities and

establish priorities, and coordinate resolution of identified problems or opportunities. Three (3) business days prior to the meeting, the Contractor shall provide the COR with an agenda of the meeting. Within three (3) business days following the meeting, the CPM shall provide to the COR written minutes of these meetings, including attendance, issues discussed, decisions made, and action items assigned. Any non-concurrence of the meeting notes content by the Government will be provided to the Contractor for revision within three (3) business days following receipt of the minutes. The Contractor shall make the necessary changes and provide them to the Government within two (2) business days.

3.10.9 Interim Progress Reviews

The Contractor shall provide the government technical lead a report at the end of each week that summarizes significant technical activities accomplished and planned technical activities for the following week.

The Contractor shall support a quarterly Program Management Review Brief, as required by the COR. The COR will provide 20 business days' notice on scheduling the IPR. For planning purposes, the IPR will be hosted alternately between the Contractor and Government sites with the first one hosted at the CONUS Contractor site. At a minimum, the IPR shall include a synopsis of activities for the quarter, accomplishments (such as releases, incidents, and problems resolved), risks, issues, and recommendations. The IPR shall also include a high-level overview of activities for the upcoming quarter. At the review, the Contractor shall develop and present a brief that addresses, at a minimum, the following areas:

- Contractor IMS to summarize the status of scheduled tasks, dependency impacts, critical path, explanation of deviations from the baseline schedule and changes to the critical path, and forecasted schedule impacts
- Contractor IMS will summarize CDRL delivery of critical plans, (e.g., Detailed Test Plan, Test Strategy, SEMP) in support of the project execution. It is required that sufficient time be available for team review and comment on the CDRLs, plans, and other key deliverables. In addition to deliverable due dates, the IMS shall indicate timelines for comment and review for CDRLs.
- Contractor metrics and corrective action plans
- Project risks, issues, and mitigations
- Current actual cost data for all task orders by CLIN (FPI and CR Only)
- System Availability trends YTD for report period
- Helpdesk metrics and trouble ticket backlog with trend analysis
- Technical progress and status on any problems reported during the preceding period
- Actual accomplishments, planned accomplishments, and status of action items
- Subcontract management

3.10.10 Meeting and Conference Support

Meeting and conference support shall include coordinating and scheduling meetings, preparing budget data, developing agenda items and risk issues, attending meetings, attending high-level meetings as required, producing meeting briefing presentations and materials, generating meeting minutes, and tracking action items.

When directed, the Contractor shall submit Meeting Minutes.

3.10.11 Key Positions

The Contractor shall provide skilled IT specialists capable of achieving program requirements and that meet the qualifications provided in the contract. The contractor shall provide Key Positions, for the following positions: Program Manager, Chief Engineer, Information System Support Lead, and Operations and Support Lead. Minimum qualifications are listed in Table 5. Any personnel that are not certified to the required standard within 30 days after award/appointment would be considered unqualified and not able to serve in a Key Position.

Key Positions support tasks and specifications prescribed in Section 3, Specific Requirements, but are ultimately accountable for all technical solutions and deliverables provided to GCSS-MC/LCM Incr 1.

KEY POSITION TABLE			
Billet Function	Education & Years of Experience (Required)	Certifications (Required/Desired)	Qualifications/Experience (Desired)
Program Manager	• Masters, Information Systems; or • Masters, Business Administration; or • BS/BA + Min 8 years' experience	• PMP (Desired) • IAM Level II (Desired) • ITILv3 Foundations (Required)	• Experience managing programs or portfolio of programs in excess of \$25M/year in revenue • Management experience in an IT environment directing 50 or more personnel. • Experience with operational and technical requirements similar to GCSS-MC/LCM Increment 1 PDSS. • 5+ years' experience in a network operations or IT operations center. • Excellent written and verbal communication skills.
Chief Engineer	• MS +10; or • BS Eng + 12; or • BS CS +12	• Oracle EBS • ITILv3Expert • IAT Level II (All required)	• 5+ years Oracle EBS experience • 5+ years technical experience in a network operations or IT operations center. • 5+ years' experience directing 50 or more technical staff members.
Information Systems Support Lead	• BS/BA +10; or • HS +15	• IAM/IAT Level III (Required)	• +7 years IA experience
Operations & Support Lead	• BS/BA +8; or • HS +12	• ITILv3 Practitioner (Required) • HDI-TSP (Required)	• Oversees responsive Service Desk operations, including service request, incident & problem response processes to maintain high customer satisfaction • Manages IT production operations to ensure system achieves established performance objectives • 7+ years' experience working in a network operations or IT operations center • 5+ years' experience implementing IT operations concepts, practices, and procedures.

Table 5: Key Positions

The Contractor shall provide personnel capable of completing the contract requirements within 30 days after contract award and for the duration of the contract. Contractors are required to follow all current and future DoD, DISA, and Marine Corps security requirements.

The Contractor shall provide personnel to support mission essential situations. On occasion, and in accordance with DoDI 3020.37, "Continuation of Essential DoD Contractor Services During Crises", services may be required to support mission essential situations (i.e., accident and rescue operations, civil disturbances, natural disasters, military contingency operations, exercises, continuing resolutions, etc.). Mission essential situations may necessitate the Contractor provide increased or reduced support beyond normal duty hours.

3.11 CONTRACT TRANSITION The overarching objective of transition is to minimize any impacts to system operations and to prevent negative impacts to GCSS-MC/LCM Increment 1 services. During the transition period, the Contractor shall ensure its personnel provide the necessary knowledge and required resources to the Government and replacement contractor to allow performance of all PDSS tasks.

3.11.1 Transition-In

If the Contract award results in selection of a new contractor, a transition period shall be required to transfer GCSS-MC incumbent contractor responsibilities to the new awardee. The transition task objective is to transfer all roles, responsibilities, and knowledge held by the incumbent contractor over to the new awardee in such ways that minimize any negative impacts to system operations and GCSS-MC Enterprise services.

For the non-incumbent, the Transition-In period is 6 months, starting upon the award of this effort. The Transition-Out period for the awardee of this Contract is 6 months and will occur on the final contract.

3.11.2

The contractor's Initial Transition-In/Migration Plan shall be delivered with the contractor's proposal in accordance with deliverable requirements in Section 5.3.1, Appendix C. The plan shall be updated to a final version no later than five days after the PAC. The contractor shall deliver weekly updates on their progress toward meeting transition criteria, and readiness to assume full support service responsibilities no later than 6 months after award. The contractor shall describe how they will reduce and mitigate potential risks. During the transition-in, the Government will provide the contractor access to the systems, databases, repositories, and facilities as required to execute transition to the contractor's full support status. This process will be managed by the Government. The Government will provide POCs for facilities operated by different organizations that might follow different access procedures. The Government will provide POCs at the PAC/Project Kick-Off (PKO) meeting. The contractor shall interface with the POCs to gain access to their systems. The contractor shall communicate all issues regarding access to any systems required in the support of this PWS to the COR within 24 hours if the issue is not already resolved. In executing the transition, program stakeholders include, but are not limited to, the LI2S-MC PMO, the incumbent contractor, hosting facilities, support vendors, and the GCSS-MC Enterprise user community. At a minimum, the contractor shall meet the transition-in criteria listed below. The contractor shall add to its plan any other criteria deemed necessary for a successful transition-in.

- Transition organization.
- Coordination of transition-out procedures with incumbent contractor.
- Staffing, including, but not limited to, use of transition personnel.
- Transition management.
- Transition communications.
- Establishing knowledge of GCSS-MC Enterprise system's applications and operations, including, but not limited to, ESD services.
- Review of GCSS-MC documentation, including, but not limited to, CRs, Backlogs, and Engineering Change Proposals (ECPs).

- Assumption of support responsibilities, including, but not limited to, operations, PROD and DEV environments, and support functions.
- Transition schedule, including, but not limited to, Plan of Action and Milestones.
- Managing transition risks and issues.
- Verification of asset management repository.
- Verification of all Government Furnished Equipment and GFI.
- Verification of current system baselines.

The Transition-In period shall begin upon award of this award. During the Transition-In period, the contractor shall acquire knowledge of, and experience with GCSS-MC Enterprise Sustainment and Transition efforts and processes sufficient to commence full performance of required services.

3.11.3 Transition-Out

The Transition-Out activities shall begin upon notification from the Contracting Officer to CACI, Inc. Federal. Upon receipt of the notification from the Contracting Officer, CACI, Inc. – Federal will comply with the approved Transition Out Plan to transition necessary knowledge, documentation, and support responsibility of GCSS-MC/LCM Increment 1 PDSS services to allow performance of all PDSS tasks. CACI, Inc. - Federal shall provide turn-over of all system documentation to ensure the replacement contractor is able to commence and perform all PDSS tasks. The Contractor shall provide documentation and assistance to ensure a seamless transition at the end of the contract.

Transition of responsibilities for full life-cycle system sustainment of GCSS-MC/LCM Increment 1 shall be executed in a phased and controlled manner that progressively transfers responsibilities from the incumbent to the new Contractor upon in accordance with the approved Transition Out Plan upon notification from the Contracting Officer.

3.13 AMERICANS WITH DISABILITIES ACT (ADA)

The U.S. Department of Defense is committed to making its electronic and information technologies accessible to individuals with disabilities in accordance with Section 508 of the Rehabilitation Act (Title 29 U.S.C. § 794d), as amended.

Within these regulations, Section 508 of the Rehabilitation Act of 1973 establishes rules to ensure information and communication technology (ICT) is accessible. Section 508 was recently refreshed, and the Revised 508 Standards are effective for federal agencies as of January 18, 2018. The GCSS-MC software shall be compliant with Section 508 of the Americans with Disabilities Act. For more information about Section 508, please visit the DoD Section 508 website (<http://dodcio.defense.gov/DODSECTION508.aspx>).

4 APPLICABLE REFERENCES, STANDARDS AND POLICIES

The following documents are applicable to this PWS and will be provided during the initial Kickoff meeting. The GCSS-MC/LCM Increment 1 tailored ITILv3® framework and process implementation plans shall align with the documents listed in Table 6.

Table 6, Reference Document List.

Reference PWS Location	Document/ Reference #	Title	Intended Use
2	DOD Manual 5200.01, Volume 4	DoD Information Security Program: Controlled Unclassified Information (CUI)	CDRL Deliverable marking guidance
2	DoD 8570.1-M	Information Assurance Workforce Improvement Program	IA/Cybersecurity certification and training requirements
2	Defense Federal Acquisition Regulation Supplement clause 252.239-7001	Information Assurance Contractor Training and Certification	IA/Cybersecurity certification and training requirements
3	Federal Acquisition Regulation (FAR) 9.505-4	Obtaining Access to Proprietary Information	Non-Disclosure Agreements
3.4.1	DoD Instruction 8500.01	Cybersecurity	Cybersecurity
3.4.1	DoD Instruction 8500.2	Information Assurance (IA) Implementation	Information Assurance
3.4.1	NIST Special Publication 800-171	Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations	Information Assurance/Cybersecurity
3.4.1	DoD DISA Document	Enclave Test and Development Security Technical Implementation Guide (STIG) Overview, Version 1, Release 3 dtd 22 January 2016	Information Assurance/Cybersecurity
3.5.3	NAVMC 1533.1	Systems Approach to Training (SAT) Manual	Training
3.5.3	MCO 1553.2B	Management of Marine Corps Formal Schools and Training Detachments	Training
3.5.3	MCO 1553.3A	Unit Training Management	Training
3.5.3	USMC TECOM College of Distance Education and Training	MarineNet Courseware Development Technical Standards, Version 2.0 dtd July 18, 2012	Training
3.5.3	DOD Instruction 1322.2	Sharable Content Object Reference Model (SCORM)	Training
3.5.3	GCSS-MC Document	eLearning Manual and Style Guide	Training
3.5.3	GCSS-MC Document	Design Standards Release 7 V2.pdf	Training
3.8.1	DoD Directive 5000.01	The Defense Acquisition System	Systems Engineering
3.8.1	MCO 5400.5	System Engineering Technical Review (SETR) Process	Systems Engineering
3.8.1	GCSS-MC Document	Sustainment Engineering Plan (SEP) Version 1, dtd 24 October 2016	Systems Engineering
3.8.4	MCO 5239.2B	Marine Corps Cybersecurity	Cybersecurity
3.8.4	DoD Instruction 8510.01	Risk Management Framework (RMF) for DoD Information Technology (IT)	Information Assurance/Cybersecurity
3.8.5	GCSS-MC-CM-009 dtd 30 December 2016	GCSS-MC/LCM Configuration Management Plan	Configuration Management

Reference PWS Location	Document/ Reference #	Title	Intended Use
3.10.2.1	USD(ATL) Memorandum dtd 9 January 2017	Department of Defense Risk, Issue, and Opportunity Management Guide for Defense Acquisition Programs	Risk Management
3.10.4	GCSS-MC Document	GCSS-MC Risk, Issue, and Opportunity Management Plan (RIOMP), Version 4, dtd 16 November 2015	Risk Management
3.10.11	DoD Instruction 3020.37	Continuation of Essential DoD Contractor Services During Crises	Continuity of Services
3.13	Rehabilitation Act (Title 29 U.S.C. § 794d), as amended	Americans with Disabilities Act Section 508	Accessibility

Table 6: Reference Document List

5 NOTIONAL DELIVERABLES/DELIVERY SCHEDULE

Section J provides a list of CDRLs. Unless specified, the Contractor shall propose a format that satisfies the CDRL in a cost effective manner for Government review and approval.

The Contractor shall provide all deliverables in Microsoft Office compatible electronic copies, unless otherwise specified (e.g., code delivery), to the COR. When approved, the Contractor should post approved product deliverables to the Government's SharePoint Online portal.

Deliverables should initially be provided in draft format for Government review/comments, except for monthly status reports, updated WBS, and CCB minutes. The Government will have thirty (30) days to review draft deliverables and provide comments for inclusion/resolution in the final versions for acceptance. Products and deliverables found "unacceptable" after final submission shall be corrected and resubmitted within ten (10) business days of Government rejection.

Hardware and software delivered should be listed in Invoice, Receipt, Acceptance and Property Transfer(iRAPT) and accepted at destination by a representative of the Government. The Contractor should prepare the 2-in-1 format in iRAPT and submit all receipt documentation to the COR.

6 GOVERNMENT FURNISHED EQUIPMENT

The Government will provide laptops to the contractor for performance in the Zone D security environment. No other GFE is currently available. The GFE to be provided to the contractor is listed in Section J, Attachment 3 - GFE. At its discretion, the Government may provide less than, but not more than, the quantities listed in the GFE Attachment.

7 ACCESS TO GOVERNMENT FACILITIES

The Government can provide access to the facilities located in Stafford, VA as needed for performance and meetings. Access to Stafford, VA is limited to meetings and short duration events and the contractor shall submit ~~JPAS~~ Defense Information System for Security (DISS) unclassified visit requests to coordinate access. No permanent workspace will be provided. All access must be coordinated in advance. Any unclassified contractor-furnished laptops must be logged with the entry control point sentry. Hours may be further reduced during the working day before and the day after Government holidays for locations not requiring 24/7 access. Access to other Government facilities will be coordinated through the program office.

8 GOVERNMENT FURNISHED MATERIALS

No Government Furnished Material will be provided under this contract.

The Contractor shall furnish their own personal computer equipment, administrative supplies, and office supplies required for the performance of the tasking within this contract.

The Government will not reimburse the Contractor for cell phone and/or air card charges under this contract.

9 GOVERNMENT FURNISHED INFORMATION (GFI)

The Government will provide access to GFI required for the performance of the contract. The Contractor needs to contact the Government, using the form found at Attachment 4, to acquire the GFI. The contractor agrees that it shall not, without authority, release, disclose, or use in any way GFI that would harm or damage the interests of the Government. Further the contractor shall not permit or result in the unauthorized disclosure or any party outside the government, or use GFI, generated during or as a result of performance of this contract or derived from GFI in any manner beyond what is necessary for the performance of this contract. GFI is not authorized for use by the contractor for the preparation of any proposal for a follow-on or other contract with the Federal Government, or for use in litigation against the Government unless the disclosure is protected by Federal Law for Whistleblowers or a similar statute. This prohibition shall expire after a period of three years after completion of performance of this contract. All contractor employees shall sign a non-disclosure agreement, establishing an enforceable agreement between the Government and the individual contractor employee acknowledging that no person shall release, disclose, or use in any way that would permit or result in disclosure to any party outside the government of any GFI or the proprietary information of any third party, that was furnished to the contractor employee for the performance of this contract or that was gained inadvertently by the employee. Further, the prohibition against release of GFI extends to cover such information whether or not in its original form, e.g., where the information has been included in contractor generated work or where it is discernible from materials incorporating or based upon such information. These prohibitions regarding GFI shall apply with equal force to any affiliate of the contractor, any subcontractor, consultant, or any entity into or with whom the contractor may merge or affiliate, or its successors or assigns and shall not expire after any given period of time.

The contractor shall immediately make full disclosure of any violation by the contractor or an employee of the contractor, whether intentional or unintentional, of the above prohibition in writing to the Government's Information Assurance Officer and the contracting officer. The notification shall include a description of the circumstances surrounding the improper disclosure of GFI, a description of the contractor's actions taken to recover, mitigate and prevent future improper disclosures, and any other relevant information.

10 TRAVEL AND OTHER COSTS (ODC)

There will be no reimbursement for travel for contractor personnel unless travel is coordinated and approved by the Government in advance of the trip. All travel requests (including purpose, dates, itinerary, estimated costs) must be submitted to the COR for approval and authorization not later than 5 business days prior to the projected travel. Upon approval, the travel costs become a "not to exceed" amount. Any incurred costs greater-than the approved "not to exceed" amount for that trip is an unallowable expense (future travel underruns will not offset exceeded travel). Upon completion of a trip, the travelers will complete and provide an expense statement to the COR. Upon approval of the submitted charges, the contractor may invoice under the travel CLIN for the costs. It is the Contractor's responsibility to include all travel costs for meetings, conferences, and kickoff required by this PWS in the firm-fixed price.

Other costs includes materials, incidentals, or specialized labor that will be necessary to fulfill work under PWS 3.7, 3.9, or 3.12. All must be approved prior to any purchase and delivery. Failure to obtain pre-authorization will make the contractor liable for the costs.

11 SECURITY REQUIREMENTS

11.1 ORGANIZATION

11.1.1 Security Classification

The information provided to the contractor will be unclassified and/or CUI. Certain contractors will be required to perform IT-I/II duties that will require favorably adjudicated Tier 5/3 Level investigations. The Defense Security Service (DSS) will not authorize Contractors to submit the necessary Tier Level investigations solely in support of IT level designation requirements, without a valid classified requirement as specified in a DD254. This effort does not warrant a DD Form 254, therefore the Government Contracting Activity Security Office (GCASO) will be required to submit any required investigations in support of IT level designations. The Contractor will be required to provide a roster of prospective contractor employees performing IT Level II and/or IT Level I duties to the MCSC COR. This roster must include: full names, Social Security Numbers, IT Level required, e-mail address and phone number. The COR will verify the IT Level requirements and forward the roster to the GCASO. Contractors found to be lacking required investigations will be contacted by the GCASO.

11.1.2 Security Officer

The Contractor shall appoint a Facility Security Officer (FSO) to support those contractor personnel requiring access to Government facility/installation and/or access to information technology systems under this contract. The FSO is the Contractor's main POC for security issues. The FSO shall have a U.S. Government security investigation that is favorably adjudicated. The FSO shall be responsible for tracking the security requirements for all personnel (subcontractors included) utilized on this contract. Responsibilities include tracking and updating the personnel security related and mandatory training information.

11.2 PERSONNEL

The Contractor shall conform to the security provisions of DoDI 5220.22/DoD 5220.22-M - National Industrial Security Program Operating Manual (NISPOM), SECNAVINST 5510.30, DoD 8570.01-M, and the Privacy Act of 1974. The Contractor shall comply with the above and the Privacy Act of 1974 to the extent that such, by their terms, are expressly applicable to the Contractor's delivery of services under this agreement and impose obligations directly upon the Contractor in its role as an information technology services provider with respect to the services performed under this agreement. The Contractor's methodology for compliance with all privacy requirements are as stated in the Contractor's Services Privacy Policy in effect as of the date of award. However, the Contractor's policy above shall in no event be construed to avoid or negate any obligation otherwise required by law. In the event of conflict, the statutory language shall prevail. Further, the Contractor's policy above regarding dispute resolution is not applicable to the Federal Government. Any dispute between the Contractor and Government will be resolved under the Contract Disputes Act and applicable clauses(s) of this contract. Prior to any labor hours being charged on contract, the contractor shall ensure all personnel (including administrative and subcontractor personnel) have obtained and can maintain favorable background investigations at the appropriate level(s) for access required for the contract/task order, and if applicable, are certified/credentialed for the Cybersecurity Workforce (CSWF). A favorable background determination is determined by either a National Agency Check with Inquiries (NACI), National Agency Check with Law and Credit (NACLC), or Single Scope Background Investigation (SSBI) and favorable Federal Bureau of Investigation (FBI) fingerprint checks. Investigations are not necessarily required for personnel performing unclassified work who do not require access to Government installations/facilities, Government IT systems and IT resources, or GCSS-MC PMO information. Cost to meet these security requirements is not directly chargeable to task order.

NOTE: If a final determination is made that an individual does not meet or cannot maintain the minimum fitness standard, the contractor shall permanently remove the individual from GCSS-MC PMO facilities, projects, and/or programs. If an individual who has been submitted for a fitness determination or security clearance is "denied" or receives an "Interim Declination," the contractor shall remove the individual from GCSS-MC PMO facilities, projects, and/or programs until such time as the investigation is fully adjudicated or the individual is resubmitted and is approved. All contractor and subcontractor personnel removed from facilities, projects, and/or programs shall cease charging labor hours directly or indirectly on task and contract.

11.2.1 Personnel Clearance

All personnel associated with this contract shall have current security investigations relating to the level of IT work being performed (IT-I, IT-II, etc.). Prior to starting work on the task, contractor personnel shall have the required clearance granted by the Department of Defense Consolidated Adjudications Facility (DoD CAF) and shall comply with IT access authorization requirements. In addition, contractor personnel shall possess the appropriate IT level of access for the respective task and position assignment as applicable per DoDI 8500.01, DoD Instruction for Cybersecurity. Any future revision to the respective directive and instruction will be applied to the TO level. Contractor personnel shall handle and safeguard any CUI in accordance with appropriate Department of Defense, Navy, and GCSS-MC PMO security regulations. The Contractor shall immediately report any security violation to the GCSS-MC PMO Security Management Office, the COR, and Government Project Manager. The Contractor shall notify the Government (written notice) within twenty-four hours of any contractor personnel added or removed from the contract that have been granted access to IT Systems, issued a Common Access Card and/or MCSC Building badge/access.

11.2.2 Access Control of Contractor Personnel

11.2.2.1 Physical Access to Government Facilities and Installations

Contractor personnel shall physically access Government facilities and installations for purposes of site visitation, supervisory and quality evaluation, work performed within Government spaces (either temporary or permanent), or meeting attendance.

- (a) All contractor persons engaged in work while on Government property shall be subject to inspection of their vehicles at any time by the Government, and shall report any known or suspected security violations to the Security Department at that location.

11.2.2.2 Identification and Disclosure Requirements

Pursuant to DFARS clause 211.106, Contractors shall take all means necessary to not represent themselves as Government employees.

11.2.2.3 IA/Cybersecurity Personnel Requirements and IT Position Categories

1. In accordance with DoD 8570.1-M, Information Assurance Workforce Improvement Program and the Defense Federal Acquisition Regulation Supplement 252.239-7001 supporting personnel must comply with certification and training requirements on the date of contract award. All new personnel must meet the certification requirements before starting work on the contract.
 - The Contractor shall ensure that personnel accessing information systems have the proper and current information assurance certification to perform information assurance functions in accordance with DoD 8570.01-M, Information Assurance Workforce Improvement Program. The Contractor shall meet the applicable information assurance certification requirements, including-
 - DoD-approved information assurance workforce certifications appropriate for each category and level as listed in the current version of DoD 8570.01-M; and
 - Appropriate operating system certification for information assurance technical positions as required by DoD 8570.01-M.
 - Upon request by the Government, the Contractor shall provide documentation supporting the information assurance certification status of personnel performing information assurance functions.
 - Contractor personnel who do not have proper and current certifications shall be denied access to DoD information systems for the purpose of performing information assurance functions.
 - The Contractor Information System Support Lead must have an IA Management Level 3 Certification in accordance with DoD 8570.-M and maintain the certification throughout the life of the contract (refer to: <http://iase.disa.mil/iawip/Pages/iabaseline.aspx>). Acceptable certifications are:
 - CISSP – Certified Information Systems Security Professional (Preferred)
 - GSLC – Global Information Assurance Certification (GIAC) Security Leadership Certificate
 - CISM - Certified Information Security Manager

- Personnel supporting IA/Cybersecurity must have the IA Technical Level 3 Certification in accordance with DoD 8750.1M and maintain the certification throughout the life of the contract (refer to <http://iase.disa.mil/iawip/Pages/iabaseline.aspx>). Acceptable certifications are:
 - CISSP – Certified Information Systems Security Professional (Preferred)
 - GCIH – Global Information Assurance Certification (GIAC) Certified Incident Handler
 - GCED – GIAC Certified Enterprise Defender
 - CISA – Certified Information Systems Auditor
 - CASP – CompTIA Advanced Security Practitioner
- Other IA personnel must have IA Management Level 2 Certification in accordance with 8570.1-M and maintain the certification throughout the life of the contract (refer to: <http://iase.disa.mil/iawip/Pages/iabaseline.aspx>). Acceptable certifications are:
 - CISSP – Certified Information Systems Security Professional (Preferred)
 - GSLC – Global Information Assurance Certification (GIAC) Security Leadership Certificate
 - CISM - Certified Information Security Manager
 - CAP – Certified Authorization Professional
 - CASP - CompTIA Advanced Security Practitioner
- Additional IA support personnel prerequisites besides the DoD 8570.01-M qualifications include the following:
 - Citizenship - All Contractor IA support personnel must be United States Citizens
 - Clearance - A Fully and Successfully Adjudicated NACLC background investigation necessary to achieve the clearance eligibility required for all GOC support and IA personnel, necessary for the performance of their duties, is required at the time of contract award. An Interim Secret Clearance eligibility or higher is required to start work. Other billets may also require a fully and successfully adjudicated background investigation and clearance eligibility as required for these positions by DoDI 8500.2, and SECNAV Manual 5510.30.

Pursuant to DoDI 8500.01, DoD Directive 8140.03, SECNAVINST 5510.30, SECNAV M-5239.20A, and applicable to unclassified DoD information systems, a designator is assigned to certain individuals that indicates the level of IT access required to execute the responsibilities of the position based on the potential for an individual assigned to the position to adversely impact DoD missions or functions. Contractor personnel shall possess the appropriate IT level of access for the task and position assignment as applicable per DoDI 8500.01, “Cybersecurity.” As defined in DoDI 5200.2 and SECNAVINST 5510.30, three basic DON IT levels/position categories are pertinent for this effort:

IT-I (Privileged Access).

IT-II (Limited Privileged, sensitive information).

IT-III (Non-Privileged, no sensitive information).

Note: The term IT Position is synonymous with the older term Automated Data Processing (ADP) Position (as used in DoD 5200.2-R, Appendix 10).

Investigative requirements for each category vary, depending on the role and whether the individual is a U.S. civilian contractor or a foreign national (foreign nationals are not allowed to access source code). The Contractor PM shall assist the Government Project Manager or COR in determining the appropriate IT Position Category assignment for all contractor personnel. All required Single-Scope Background Investigation (SSBI), SSBI Periodic Reinvestigation (SSBI-PR), and National Agency Check (NAC) adjudication will be performed Pursuant to DoDI 8500.01 and SECNAVINST 5510.30. Requests for investigation of contractor personnel for fitness determinations or IT eligibility without classified access are submitted by GCSS-MC PMO Security Office, processed by the OPM, and adjudicated by DOD CAF. IT Position Categories are determined based on the following criteria:

11.2.2.3.1 IT-I Level (Privileged Access)

IT-I Level privileged roles are organization-defined roles assigned to individuals that allow those individuals to perform certain security-relevant functions that ordinary users are not authorized to perform. Privileged roles include key management, account management, database administration, system and network administration, and web administration. A role-based access scheme organizes permitted system access and privileges into roles. In contrast, an attribute-based access scheme specifies allowed system access and privileges based on attributes.

11.2.2.4 IT-II Level (Limited Privileged)

Positions in which the Contractor is responsible for the-direction, planning, design, operation, or maintenance of a computer system, and whose work is technically reviewed by a higher authority at the IT-II Position level to insure the integrity of the system. Personnel whose duties meet the criteria for an IT-II Position require a favorably adjudication of a Position of Trust National Agency Check with Law and Credit (PT/NACLC). Assignment to designated IT-II positions requires U.S. citizenship unless a waiver request is approved by CNO.

11.2.2.5 IT-III Level (Non-privileged)

All other positions involved in computer activities. The Contractor in this position has non-privileged access to one or more DoD information systems/applications or database to which they are authorized access. Personnel whose duties meet the criteria for an IT-III Position designation require a favorably adjudication of a Position of Trust National Agency Check with Written Inquiries (PT/NACI).

11.2.3 Security Training

Regardless of the contract security level required, the Contractor shall be responsible for verifying applicable personnel (including subcontractors) receive all required training. At a minimum, the Contractor's designated Security Officer shall track the following information: security clearance information; dates possessing CACs; issued and expired dates GCSS-MC PMO Badge; Cybersecurity training; Privacy Act training; Personally Identifiable Information (PII) training; Cybersecurity Workforce (CSWF) certifications; etc. The contractor shall educate employees on the procedures for the handling and production of classified material and documents, and other security measures as described in the PWS in accordance with DoD 5220.22-M.

11.2.4 Disclosure of Information

In support of DFARS Clause 252.204-7000, contractor employees shall not discuss or disclose any information provided to them in the performance of their duties to parties other than authorized Government and contractor personnel who have a "need to know". The Contractor shall not use any information or documentation developed by the contractor under direction of the Government for other purposes without the consent of the Government Contracting Officer.

11.2.5 Handling of Personally Identifiable Information (PII)

When a contractor, including any subcontractor, is authorized access to Personally Identifiable Information (PII), the contractor shall complete the annual PII training requirements, found at <https://twms.nmci.navy.mil>, and comply with all privacy protections under the Privacy Act (Clause 52.224-1 and 52.224-2). The contractor shall safeguard PII from theft, loss, and compromise. The contractor shall transmit and dispose of PII in accordance with the latest DON policies. The contractor shall not store any Government PII on their personal computers. The contractor shall mark all developed documentation containing Personally Identifiable Information (PII) accordingly in either the header or footer of the document: 'CUI – Privacy Sensitive. Any misuse or unauthorized disclosure may result in both criminal and civil penalties.' Any unauthorized disclosure of privacy sensitive information through negligence or misconduct can lead to contractor removal or contract termination depending on the severity of the disclosure. Upon discovery of a PII breach, the contractor shall immediately notify the Contracting Officer and COR. Contractors responsible for the unauthorized disclosure of PII shall be held accountable for any costs associated with breach mitigation, including those incurred as a result of having to notify personnel. The Contractor shall comply with all Privacy Act requirements to the extent that by its terms, is expressly applicable to the Contractor's delivery of services under this agreement and imposes obligations directly upon the Contractor in its role as an information

technology services provider with respect to the services performed under this agreement. The Contractor's methodology for compliance with all privacy requirements are as stated in the Contractor's Services Privacy Policy in effect as of the date of award. However, the Contractor's policy above shall in no event be construed to avoid or negate any obligation otherwise required by law. In the event of conflict, the statutory language shall prevail. Further, the Contractor's policy above regarding dispute resolution is not applicable to the Federal Government. Any dispute between Contractor and Government will be resolved under the Contract Disputes Act and applicable clause(s) of this contract.

11.3 OPERATIONS SECURITY (OPSEC) REQUIREMENTS

Security programs are oriented towards protection of classified information and material. OPSEC is an operations function which involves the protection of any critical information – focusing on unclassified information that may be susceptible to adversary exploitation. Pursuant to DoDD 5205.02E and GCSS-MC PMO's OPSEC program implements requirements in DoD 5205.02-M – OPSEC Program Manual. Note: OPSEC requirements are applicable when contract personnel have access to classified information or unclassified Critical Program Information (CPI)/sensitive information.

11.3.1 Local and Internal OPSEC Requirement

Contractor personnel, including subcontractors if applicable, shall adhere to the OPSEC program policies and practices as cited in the GCSS-MC PMO Instruction as well as existing local site OPSEC procedures. The contractor shall develop their own internal OPSEC program specific to the contract and based on GCSS-MC PMO OPSEC requirements. At a minimum, the Contractor's program shall identify the current GCSS-MC PMO site OPSEC Officer/Coordinator.

11.3.2 OPSEC Training

The Contractor shall track and ensure applicable personnel receive initial and annual OPSEC awareness training. Training may be provided by the Government or a Contractor's OPSEC Manager. Contractor training shall, as a minimum, cover OPSEC as it relates to contract work, discuss the Critical Information applicable in the contract/task order, and review OPSEC requirements if working at Government facilities. The Contractor shall ensure any training materials developed by the Contractor shall be reviewed by the GCSS-MC PMO OPSEC Officer, who will ensure it is consistent with GCSS-MC PMO OPSEC policies. OPSEC training requirements are applicable for personnel during their entire term supporting GCSS-MC PMO contracts.

11.3.3 GCSS-MC PMO OPSEC Program

Contractor shall participate in GCSS-MC PMO OPSEC program briefings and working meetings, and the Contractor shall complete any required OPSEC survey or data call within the timeframe specified.

11.4 DATA HANDLING AND USER CONTROLS

11.4.1 Data Handling

The Contractor shall handle all data received or generated under this contract as CUI material unless otherwise marked. Public release of unclassified, uncontrolled DoD information accessed under this contract shall be approved by the Contracting Officer, after it has been routed through the MCSC Security, Public Affairs and OPSEC representatives. The Contractor shall handle all information received or generated in accordance with its classification and be in compliance with all required PWS references and other required Government policies and procedures that include DoD/Navy/GCSS-MC PMO in the text of a Task Order.

11.4.2 Effective Use of Controls

The Contractor shall screen all electronic deliverables or electronically provided information for malicious code using DoD approved anti-virus software prior to delivery to the Government. The Contractor shall utilize appropriate controls (firewalls, password protection, encryption, digital certificates, etc.) at all times to protect contract related information processed, stored or transmitted on the contractor's and Government's computers/servers to ensure confidentiality, integrity, availability, authentication and non-repudiation. The contractor shall ensure provisions are in place that will safeguard all aspects of information operations pertaining to

this contract in compliance with all applicable PWS references. The contractor shall ensure Data-at-Rest is required on all portable electronic devices including storage of all types. Encryption/digital signing of communications is required for authentication and non-repudiation.